

Self-gravity versus phase mixing at fixed angular momentum

Linear theory and the design of a particle-in-cell test battery

Working notes

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Abstract

We study whether, in the reduction $f = F(r, p_r) \delta(L - L_0)$ with an isochrone background, the self-gravity of a component of small mass can sustain an undamped oscillation, and what happens near the transition between damping and oscillation. The coupling is measured by the ratio η between the self-gravitational shift of the orbital frequencies and the width of the frequency band. The linear problem is solved with the linearized solver of the repository. For a lowered Maxwellian (the Wilson model) with the reference width, the collective mode approaches the lower edge of the band as η increases, and its damping rate falls tenfold between $\eta = 0.45$ and 1; between $\eta = 1.0$ and 1.24 it leaves the band as an undamped discrete mode. η orders the transition, and changing L_0 with the width scaled accordingly does not move it, but the threshold depends on the width of the support, the shape of the distribution and the exponent of the vacuum edge. With a sharp edge, King included, the discrete mode is already present at $\eta = 1$, and the edge response persists at $\eta = 0.1$, in the direction of the dichotomy proved by Hadžić and collaborators. The exact criterion is the gain of the self-consistent loop at the lower edge of the band, λ_{edge} : in the 25 equilibria with compact support the mode is discrete if and only if $\lambda_{\text{edge}} > 1$, whatever the width, shape or edge, and the root of $\lambda(\omega) = 1$ gives its frequency. With a sharp edge the discrete mode exists at weak coupling, but so close to the edge that no run can separate it. Near the transition the linear damping rate tends to zero while the bounce frequency of trapped particles does not, so O’Neil’s parameter $\nu = \omega_b/\gamma$ is large for any finite amplitude and the late-time dynamics is nonlinear: at $\eta = 1$, $\varepsilon \approx 0.008$ already gives $\nu = 1$. We propose a battery of 30 particle-in-cell (PIC) runs, about 4.5 h with four threads (a core of 12 in about 2 h), organized in the (η, ν) plane, with the linear solution of each equilibrium as reference. All numbers are computed with the scripts of the repository, not estimated.

1 Theoretical framework

1.1 The fixed-angular-momentum model

We use units with $G = 1$ in which the background is an isochrone of unit mass and unit scale b , so that masses are measured in units of the background mass and times in units of $(b^3/GM_{\text{iso}})^{1/2}$. The self-gravitating component has mass a_0 ; $a_0 = 0.1$ is one tenth of the total mass of the background.

In spherical symmetry the distribution function depends on position and velocity only through the radius r , the radial velocity p_r and the modulus L of the angular momentum. The code solves the reduction

$$f(r, p_r, L) = F(r, p_r) \delta(L - L_0),$$

in which all particles have the same angular momentum L_0 . The radial motion of each particle is then one-dimensional, with Hamiltonian

$$H = \frac{1}{2}p_r^2 + \Phi_{\text{ef}}(r), \quad \Phi_{\text{ef}}(r) = \Phi(r) + \frac{L_0^2}{2r^2},$$

where $\Phi = \Phi_{\text{iso}} + \Phi_{\text{self}}$ is the total potential: the isochrone background, $\Phi_{\text{iso}} = -1/(1 + \sqrt{1 + r^2})$, plus the potential of the distribution itself. The term $L_0^2/2r^2$ is the centrifugal barrier.

The measure $8\pi^2 L_0$. In spherical symmetry the phase-space volume element is written in terms of r , p_r and L . The position contributes $4\pi r^2 dr$. The velocity has a radial component p_r and a tangential component v_t , which can point in any direction of the plane perpendicular to $\hat{r}$, so $d^3v = 2\pi v_t dv_t dp_r$. Since $L = rv_t$, $v_t dv_t = L dL/r^2$, and

$$d^3x d^3v = 4\pi r^2 dr \cdot \frac{2\pi L dL}{r^2} dp_r = 8\pi^2 L dL dr dp_r.$$

The delta function in L removes dL and leaves $8\pi^2 L_0 dr dp_r$. The mass of the component is $M = 8\pi^2 L_0 \iint F dr dp_r$.

The reduction is not an approximation within the model, but it is an extremely anisotropic model. Everything below holds for it, and comparisons with isotropic systems require care. The comparison with Hadžić and collaborators (Section 1.8) is not affected: their theorem is formulated in exactly this reduction, with all particles at the same $|L|$ in the field of a central point mass.

1.2 Angle–action variables

Since H does not depend on time and the radial motion is bounded between two turning points $r_- < r_+$, each orbit is a closed curve in the (r, p_r) plane, and angle–action variables exist. The radial *action* is the area enclosed by the orbit divided by 2π ,

$$J = \frac{1}{2\pi} \oint p_r dr = \frac{1}{\pi} \int_{r_-}^{r_+} \sqrt{2(E - \Phi_{\text{ef}}(r))} dr,$$

and the *angle* Q is the phase along the orbit, normalized so that it advances uniformly from 0 to 2π in one radial period. The map $(r, p_r) \rightarrow (Q, J)$ is canonical, so it preserves area: $dr dp_r = dQ dJ$. Since H depends only on J , Hamilton’s equations are $\dot{J} = 0$ and $\dot{Q} = \partial H / \partial J$:

$$\Omega(J) = \frac{dE}{dJ} = \frac{2\pi}{T_r(J)}, \quad T_r = 2 \int_{r_-}^{r_+} \frac{dr}{\sqrt{2(E - \Phi_{\text{ef}})}},$$

with T_r the radial period. Only the isochrone has a closed form:

$$E(J) = -\frac{1}{2(J+c)^2}, \quad \Omega(J) = \frac{1}{(J+c)^3}, \quad \frac{d\Omega}{dJ} = -\frac{3}{(J+c)^4},$$

$$c = \frac{1}{2} \left(L_0 + \sqrt{L_0^2 + 4} \right).$$

Two consequences are used throughout. The frequency *decreases* with the action: nearly circular orbits ($J \rightarrow 0$) are the fastest, and $\Omega(0) = c^{-3}$ is the highest frequency. And the derivative does not vanish, so each frequency corresponds to a single action. With self-gravity there is no closed form, and the map is computed numerically (Section 2.1).

The circular orbit and the epicycles. The orbit with $J = 0$ is circular. It lies at the minimum of Φ_{ef} , at the radius r_c where $\Phi'_{\text{ef}}(r_c) = 0$, that is, $\Phi'(r_c) = L_0^2/r_c^3$, and has energy $E_c = \Phi_{\text{ef}}(r_c)$. With $L_0 = 2$ and no self-gravity, $r_c = 5.742$.

Approximation 1: the epicycle. Near the minimum, Φ_{ef} is a parabola and the radial motion is a harmonic oscillator:

$$H \approx E_c + \frac{1}{2} p_r^2 + \frac{1}{2} \kappa^2 (r - r_c)^2, \quad \kappa^2 = \Phi''_{\text{ef}}(r_c) = \Phi''(r_c) + \frac{3L_0^2}{r_c^4}.$$

κ is the *epicyclic frequency*. The action of an oscillator is its energy divided by its frequency, so $E \approx E_c + \kappa J$ and $\Omega(0) = \kappa$. The orbit is

$$r - r_c = R_e(J) \cos Q, \quad p_r = -\kappa R_e(J) \sin Q, \quad R_e(J) = \sqrt{2J/\kappa},$$

where R_e , the epicycle radius, is the radial half-width of the orbit. (In this notation $Q = 0$ is the apocentre. The scripts and the code measure Q from pericentre, so their angle is this one plus π ; the shift does not affect any of the arguments below.) For the isochrone with $L_0 = 2$, $\kappa = \Omega(0) = c^{-3} = 0.0711$; Section 1.7 checks it against the density of the background.

Smooth coordinates near the circular orbit. With $x = \sqrt{\kappa}(r - r_c)$ and $y = -p_r/\sqrt{\kappa}$, the orbit is

$$x + iy = \sqrt{2J} e^{iQ}, \quad J = \frac{1}{2}(x^2 + y^2) :$$

$\sqrt{2J}$ is a radius and Q a polar angle in the (x, y) plane, and the circular orbit is the origin. The functions x , y and J are smooth in (r, p_r) , but Q is not defined at the origin. A function of the form $a(J) e^{\pm iQ}$ is therefore smooth at the circular orbit only if $a(J) = \sqrt{J}$ times a smooth function of J , because $\sqrt{J} e^{iQ} \propto x + iy$ is smooth and $e^{iQ} = (x + iy)/\sqrt{x^2 + y^2}$ is not. In general, $e^{\pm ikQ}$ requires a factor $J^{k/2}$.

1.3 Equilibria

In angle-action variables, with a Hamiltonian that depends only on J , the Vlasov equation is $\partial_t F + \Omega(J) \partial_Q F = 0$, so any F that depends only on J is stationary. But J is defined in the total potential, and the potential depends on F : the equilibrium must be *self-consistent*. Its construction is described in Section 2.1.

The equilibrium must depend only on J for two reasons. A distribution that depended on Q would wind up by phase mixing from $t = 0$, and that decay would be indistinguishable from the response to be measured. And without a well-defined equilibrium there is no fixed frame in which to measure $\delta\Phi$, and neither the damping rate nor the frequency of the mode would have a meaning.

Normalization. We write $F_{\text{eq}} = A \tilde{F}(J)$, with $\tilde{F}$ the shape and A the amplitude. Since $dr dp_r = dQ dJ$ and $\int dQ = 2\pi$, the mass is

$$a_0 = 8\pi^2 L_0 \iint F_{\text{eq}} dQ dJ = 16\pi^3 L_0 A \int \tilde{F}(J) dJ,$$

which fixes A . If the shape is written in the energy, as for the lowered Maxwellian, $\tilde{F}$ as a function of J changes with the potential, and A is recomputed at each iteration.

Three families. We consider three shapes, all functions of J (or of E) only. Section 4.1 explains the choice between them and of their parameters.

- **The Gaussian**, $\tilde{F} = J^2 e^{-J^2/\sigma_J^2}$, the one of the Landau runs. Its only action scale is σ_J : with $u = J/\sigma_J$, $\tilde{F} = \sigma_J^2 u^2 e^{-u^2}$, and changing σ_J only stretches the curve. It has no edge, so its support is defined by a threshold: where $\tilde{F}$ exceeds 1% of its maximum, which is e^{-1} at $u = 1$. Solving $u^2 e^{-u^2} = 0.01 e^{-1}$ gives $u = 0.061$ and $u = 2.763$, so the support is $[0.061, 2.763] \sigma_J$ and its width $\Delta J \approx 2.70 \sigma_J$.
- **The hollow family**, $\tilde{F} = J^m (J_t - J)^g$ on $[0, J_t)$ and zero outside (---**forma politropo**). It has compact support $[0, J_t]$, vanishes as J^m at the circular orbit and as $(J_t - J)^g$ at the vacuum edge, and has its maximum at $J = m J_t / (m + g)$. We use $m = g = 2$, $J^2 (J_t - J)^2$, with its maximum at $J_t/2$. With $m = 0$ it is the polytrope in the action.

- **The lowered Maxwellian** (`--forma maxwell`), written in the energy:

$$F_{\text{eq}} = A E_{\gamma}(g, x), \quad x = \frac{E_t - E}{T} \text{ if } E < E_t,$$

$$E_{\gamma}(g, x) = e^x P(g, x) = \sum_{n \geq 0} \frac{x^{g+n}}{\Gamma(g+n+1)},$$

with $P(g, x)$ the regularized lower incomplete gamma function. E_{γ} has two limits that explain its use. Near the edge, $x \rightarrow 0$ and $E_{\gamma} \approx x^g / \Gamma(g+1)$, so F vanishes as $(E_t - E)^g$, with exponent g . Far from the edge, $x \gg g$ and $E_{\gamma} \rightarrow e^x \propto e^{-E/T}$, a Maxwell–Boltzmann distribution. The King model is $g = 1$, $e^x - 1$, and the Wilson model $g = 2$, $e^x - 1 - x$. The parameters are $E_t = E(J_t)$, the energy of the orbit of action J_t , and $T = (E_t - E_c)/W_0$, with W_0 the depth of the well in units of the temperature. E_t , T and A are recomputed in the potential of each iteration, so the support is always $[0, J_t]$ and the mass a_0 . Since $E \approx E_c + \kappa J$ near the circular orbit, $x \approx W_0(1 - J/J_t)$ to first order. The series of E_{γ} has positive terms, so it is evaluated without cancellations, also at the edge.

Equivalent widths. The edge of the compact families is placed at the upper end of the support of the Gaussian, $J_t = 2.763 \sigma_J$, so that without self-gravity all families have the same frequency band. The reference width is $\sigma_J = 0.05$, or $J_t = 0.138$.

1.4 Perturbations

The initial condition is the equilibrium plus a perturbation in angle,

$$F = F_{\text{eq}}(J) [1 + \varepsilon s(J) \chi(Q)].$$

Angular profile $\chi(Q)$. It must be periodic with zero mean, so that the integral of F over Q , and with it the mass and the equilibrium, do not change. Two profiles are used (`--gq`):

- **cos:** $\chi = \cos Q$. It excites only $k = 1$. In the linear regime each harmonic evolves independently, so exciting a single one gives the cleanest measurement. It is used unless stated otherwise.
- **suma3:** $\chi = \cos Q + \frac{1}{2} \cos 2Q + \frac{1}{3} \cos 3Q$. It excites $k = 1, 2, 3$ and is used for the *common-frequency* test, the signature of a genuine mode (Section 4.4).

Radial factor $s(J)$. Three factors are used (`--pert`): $s = 1$ (**plana**), $s = \sqrt{J/J_{\text{max}}}$ (**suave**) and $s = (J/J_{\text{max}})^{3/2}$ (**suave3**). The choice is set by regularity at the circular orbit (Section 1.2):

- If $F_{\text{eq}}(0) > 0$, as for the Maxwellian, the flat perturbation is discontinuous: $F_{\text{eq}}(0) \cos Q = F_{\text{eq}}(0) x / \sqrt{x^2 + y^2}$ jumps at the origin. With $s = \sqrt{J/J_t}$ it becomes $\propto x$, which is smooth; with $s = (J/J_t)^{3/2}$ it becomes $\propto Jx$, smooth and smaller near the circular orbit.
- If $F_{\text{eq}} \propto J^2$ near $J = 0$, as for the Gaussian and the hollow family, $F_{\text{eq}} \cos Q \propto J^{3/2}x$ is already continuous, and the flat perturbation suffices.

The Maxwellian uses **suave3**, and the hollow family and the Gaussian use **plana**. The tails computed in Section 3.3 show why **suave3** rather than **suave**.

Positivity. Since $s \leq 1$, $F \geq 0$ requires $\varepsilon \max |\chi| \leq 1$: $\varepsilon \leq 1$ with **cos** and $\varepsilon \leq 0.67$ with **suma3**, whose maximum is ≈ 1.5 . The generator checks this and aborts if it fails.

1.5 Observables

We write $F = F_{\text{eq}} + \delta F$ and $\Phi = \Phi_{\text{eq}} + \delta\Phi$.

Potential perturbation. On the radial grid of the code,

$$\delta\Phi(r, t) = \Phi(r, t) - \Phi_{\text{iso}}(r) - \Phi_{\text{self,eq}}(r),$$

computed by `landau_analisis.py`. It does not depend on any angle-action map, it is what enters the equations of motion, and it is the quantity whose decay the theorems address. Its size is summarized by $\|\delta\Phi\|(t)$, the rms value over $r \in [3, 15]$, the region where the component with $L_0 = 2$ lives. As in the Landau runs, the part due to the perturbation is isolated by subtracting a run with $\varepsilon = 0$ of the same equilibrium: $[\delta\Phi(\varepsilon) - \delta\Phi(0)]/\varepsilon$.

Harmonics. The angular harmonics of the distribution, weighted by a test function of the action, are

$$h_k(t) = \frac{\sum_j f_j B(J_j) e^{-ikQ_j}}{[\sum_j f_j B(J_j)]_{t=0}}, \quad B(J) = J^2 e^{-(J-j_1)^2/s_{j_1}^2},$$

with the sum over the particles, f_j their weights and (Q_j, J_j) from the map of the equilibrium potential (`landau_analisis.py`). We record $k = 0 \dots 4$; h_0 is conserved by construction and serves as normalization. The comparison with linear theory uses $[h_k(\varepsilon) - h_k(0)]/\varepsilon$, and `lineal.py` gives its linear analogue, $\sum \delta F B e^{-ikQ} / \sum F_{\text{eq}} B$.

The test function $B(J)$ is a choice of the code (`analysish.f90`). Its centre and width follow the support, $j_1 = s_{j_1} = J_t/2.763$, the same proportion as in the Landau runs (0.10 on a support that ends at 0.276). They are set in the `.par` file of each run, from which `landau_analisis.py` and, with `--j1 --sj1`, `lineal.py` read them. The factor J^2 makes B vanish at the circular orbit; the consequences are given in Section 1.6.

The Fortran code also computes an h_k (`analysish.f90`), with an additional factor in Q , $A(Q) = e^{-\sin^2(Q/2)/s_Q^2}$, and with the map of the *bare* isochrone. The factor does not convolve the harmonics. The code defines h_k as the projection of f on harmonic k of the test function $A(Q)B(J)$, $\sum_j f_j B(J_j) C_k e^{-ikQ_j}$ with $C_k = \langle A(Q) \cos kQ \rangle$ computed once by quadrature, so each harmonic is multiplied by the constant C_k . (Multiplying each particle by $A(Q_j)e^{-ikQ_j}$ would mix harmonics, but that is not what `analysish.f90` does.) With $a_0 > 0$ the (Q, J) of that map are not the true ones, so the battery uses the h_k of `landau_analisis.py`.

General linear observables. $\delta\Phi$ and h_k are two cases of the same construction. Any quantity linear in the perturbation is the integral of δF over phase space, weighted by a fixed function $\beta(Q, J)$ that defines the observable,

$$O(t) = \iint dQ dJ \beta(Q, J) \delta F(Q, J, t),$$

and all that matters of β for the $k = 1$ harmonic are its angular Fourier coefficients along each orbit,

$$\beta_{\pm 1}(J) = \frac{1}{2\pi} \int dQ \beta(Q, J) e^{\pm iQ}.$$

- **$\delta\Phi$ at radius r .** By Poisson's equation in spherical symmetry, $\delta\Phi(r) = \int G(r, r') \delta\rho(r') d^3r'$, with $G(r, r') = -1/\max(r, r')$ the Green's function of spherical shells: a shell at r' contributes $-1/r'$ inside it and $-1/r$ outside. The density follows from integrating δF with the measure of Section 1.1, and in angle-action variables the weight is the Green's function evaluated where the particle is: $\beta(Q, J) = 8\pi^2 L_0 G(r, r(Q, J))$. No test function is involved.

- h_1 . $\beta = B(J) e^{-iQ}$ divided by the normalization, so $\beta_1 = B$ (up to the constant) and $\beta_{-1} = 0$.

By Section 1.2, a weight that is continuous at the circular orbit has $\beta_{\pm 1} \propto \sqrt{J}$ times a smooth function. So $\beta_{\pm 1} \sim J^{1/2}$ is the generic case, and $\delta\Phi$ satisfies it (Section 1.6). A β_1 that vanishes faster, like the J^2 of B , is a choice.

1.6 The frequency band and phase mixing

Without self-gravity each particle rotates in the equilibrium potential at its own frequency. Self-gravity is measured as a deviation from this free evolution.

The band. The support of F_{eq} occupies an interval of actions $[J_{\text{lo}}, J_{\text{hi}}]$ and, through $\Omega(J)$, a band of frequencies. Since Ω decreases with J ,

$$\Omega_{\min} = \Omega(J_{\text{hi}}), \quad \Omega_{\max} = \Omega(J_{\text{lo}}), \quad \Delta\Omega = \Omega_{\max} - \Omega_{\min}.$$

With compact support the band is exactly $[\Omega(J_t), \Omega(0)]$. It is the set of frequencies at which some particle can resonate, and in the linearized problem it is the *essential spectrum* (Section 1.7): a mode with frequency inside the band exchanges energy with the resonant particles and is damped, and one with frequency outside is not. We also use the mass-weighted mean frequency

$$\bar{\Omega} = \frac{\int F_{\text{eq}}(J) \Omega(J) dJ}{\int F_{\text{eq}}(J) dJ},$$

since the mass between J and $J + dJ$ is proportional to $F_{\text{eq}}(J) dJ$ (the measure is $dQ dJ$).

Approximation 2: where the support ends. With compact support there is nothing to choose. For the Gaussian the support is set by the 1 % threshold of Section 1.3. This is a convention: another threshold changes the edges and, with them, the absolute value of η , but not the comparisons between cases.

Approximation 3: the relative width to first order. If the interval of actions ΔJ is small compared with $J + c$, a Taylor expansion of $\Omega = (J + c)^{-3}$ gives

$$\frac{\Delta\Omega}{\Omega} \approx \frac{|d\Omega/dJ| \Delta J}{\Omega} = \frac{3 \Delta J}{J + c}.$$

The width of the band therefore grows with the width of the support in action ($\Delta J \approx 2.70 \sigma_J$, or J_t) and decreases with L_0 , which sets c . Section 3.2 compares this formula with the exact bands.

The mixing time. Two particles at the ends of the band accumulate a phase difference $k \Delta\Omega t$ in harmonic k . When this difference reaches 2π , the phases of the harmonic cover the whole circle. This defines

$$\tau_k = \frac{2\pi}{k \Delta\Omega},$$

the time over which phase mixing of harmonic k is complete.

The free solution. Without self-gravity the equation $\partial_t F + \Omega(J) \partial_Q F = 0$ is solved by characteristics: each point keeps its action and its angle advances uniformly,

$$F(Q, J, t) = F(Q - \Omega(J)t, J, 0).$$

For the perturbation of Section 1.4 with $\chi = \cos Q$,

$$\delta F(Q, J, t) = \varepsilon F_{\text{eq}}(J) s(J) \cos(Q - \Omega(J)t).$$

Substituting in a linear observable and integrating first over Q , with $\cos x = (e^{ix} + e^{-ix})/2$, only the coefficients $\beta_{\pm 1}$ survive:

$$O(t) = \pi\varepsilon \int dJ F_{\text{eq}}(J) s(J) \left[\beta_1(J) e^{-i\Omega(J)t} + \beta_{-1}(J) e^{i\Omega(J)t} \right].$$

For a real weight $\beta_{-1} = \beta_1^*$ and $O = 2\pi\varepsilon \text{Re} \int dJ \beta_1 F_{\text{eq}} s e^{-i\Omega t}$. For h_1 , $\beta_1 = B$ and

$$h_1(t) = \frac{\varepsilon}{2} \frac{\int dJ B F_{\text{eq}} s e^{-i\Omega(J)t}}{\int dJ B F_{\text{eq}}}.$$

All the time dependence is in $e^{-i\Omega(J)t}$; the observable enters only through $\beta_1(J)$, which says how much each orbit weighs in it.

The frequency distribution. Since $\Omega(J)$ is monotonic we can change variables, $dJ = d\Omega/|d\Omega/dJ|$:

$$O(t) \propto \int d\Omega w(\Omega) e^{-i\Omega t}, \quad w(\Omega) = \frac{\beta_1 F_{\text{eq}} s}{|d\Omega/dJ|} \Big|_{J(\Omega)}.$$

$w(\Omega) d\Omega$ is the weight, in that observable, of the orbits with frequency between Ω and $\Omega + d\Omega$. The observable is the Fourier transform of w , and the question of how it decays becomes the question of how the Fourier transform of w decays.

$O(t)$ is a sum of phasors $e^{-i\Omega t}$, one per orbit, weighted by w . At $t = 0$ they all point the same way. Each then rotates at its own rate, the phases spread over a range $\Delta\Omega t$, and when this range greatly exceeds 2π the phasors of neighbouring frequencies cancel. By the Riemann–Lebesgue lemma, if w is integrable its transform tends to zero. This is *phase mixing*, and it occurs without dissipation: the information is not lost but moves to ever finer scales in J .

Regularity of w and the form of the decay. The cancellation is complete where w is smooth, because each phasor has nearly equal neighbours on both sides. What survives comes from the points where w is not smooth.

- **w smooth and vanishing with all its derivatives at the ends.** Each integration by parts gains a factor $1/t$, so the transform decays faster than any power. If w is approximately Gaussian with standard deviation σ_Ω , its transform is $|O(t)| \approx |O(0)| e^{-\sigma_\Omega^2 t^2/2}$: Gaussian decay in time.
- **An edge.** If near $\Omega_{\min}$, with $u = \Omega - \Omega_{\min} \geq 0$, $w \approx c u^\alpha$ and $w = 0$ below, the substitution $v = ut$ gives

$$\int_0^\infty u^\alpha e^{-iut} du = t^{-(\alpha+1)} \int_0^\infty v^\alpha e^{-iv} dv = \Gamma(\alpha+1) e^{-i\pi(\alpha+1)/2} t^{-(\alpha+1)},$$

so the edge contributes a tail $c \Gamma(\alpha+1) e^{-i\Omega_{\min} t} (it)^{-(\alpha+1)}$. It oscillates at the frequency of the edge and decays as a power. At time t only phasors separated by more than $\sim 1/t$ cancel each other; those within $1/t$ of the edge have no partner on one side and weigh $\int_0^{1/t} u^\alpha du \propto t^{-(\alpha+1)}$. The sharper the edge, the slower the tail: a jump ($\alpha = 0$) leaves $1/t$.

- **An interior point** where $w \sim |\Omega - \Omega_*|^\alpha$, or where w changes form abruptly, contributes in the same way a tail $t^{-(\alpha+1)}$ at frequency Ω_* .

Near any point, $\Omega - \Omega_e \approx \Omega'(J_e) (J - J_e)$ with $\Omega' \neq 0$, so a power in J is the same power in Ω , and $|\Omega'|$ enters only the amplitude. (If Ω' vanished somewhere, a stationary frequency, there would be a stationary-phase contribution $\propto t^{-1/2}$; this does not happen in the isochrone.) This gives the rule used throughout.

Tail rule. Each point J_e where the integrand $\beta_1 F_{\text{eq}} s$ behaves as $|J - J_e|^\alpha$, whether an edge of the band or an interior point, contributes a tail $\propto t^{-(\alpha+1)}$ at frequency $\Omega(J_e)$. Where the integrand is smooth the decay is faster than any power. The point with the smallest α dominates.

Approximation 4: τ_k is a scale, not a law. The actual decay is not exponential with rate $1/\tau_k$; its form is given by the tail rule. τ_k gives the order of magnitude of the time at which mixing is complete. For the Gaussian, w is approximately Gaussian, and the 1 % threshold lies where $e^{-u^2/2} = 0.01$, that is, 3.03σ from the centre, so $\Delta\Omega \approx 6.1 \sigma_\Omega$. Section 3.3 tests this with a free run.

Tails of $\delta\Phi$. For $\delta\Phi$ the weight is $\beta = 8\pi^2 L_0 G(r, r(Q, J))$, and the free solution gives

$$\delta\Phi(r, t) = 8\pi^2 L_0 \varepsilon \operatorname{Re} \int dJ F_{\text{eq}}(J) s(J) C(r, J) e^{-i\Omega(J)t}, \quad C(r, J) = \int dQ e^{iQ} G(r, r(Q, J)).$$

$C(r, J)$ is the contribution of the $k = 1$ harmonic of orbit J to the potential at r . Three properties, which follow only from the orbits and from Poisson's equation, set the tail:

- **Circular orbit.** Near it the orbit is an epicycle, $r(Q, J) \approx r_c + R_e(J) \cos Q$. Expanding G about r_c , $G(r, r_c + R_e \cos Q) \approx G(r, r_c) + \partial_{r'} G R_e \cos Q$, and the Q integral of the constant term vanishes, so $C \approx \pi \partial_{r'} G R_e \propto \sqrt{J}$. The end $J = 0$ of the band (Ω_{max}) contributes an integrand $\propto F_{\text{eq}} s J^{1/2}$. If $F_{\text{eq}}(0) > 0$, as for any monotonic distribution, the tail is $t^{-3/2}$ with $s = 1$, t^{-2} with $s = \sqrt{J/J_t}$ and t^{-3} with $s = (J/J_t)^{3/2}$; if $F_{\text{eq}} \propto J^2$, it is $t^{-7/2}$.
- **Vacuum edge.** C does not vanish at J_t , so if $F_{\text{eq}} \propto (J_t - J)^g$ the tail is $t^{-(g+1)}$, at Ω_{min} .
- **Turning points.** $G(r, r')$ has a kink at $r' = r$: it is $-1/r$ for $r' < r$ and $-1/r'$ for $r' > r$. If the apocentre $r_+(J)$ of the orbit lies just above r , the arc of the orbit beyond r has a half-width in Q proportional to $\sqrt{r_+ - r}$ (near apocentre, $r_+ - r(Q)$ grows as the square of the distance in Q), and on it G differs from $-1/r$ by $\approx (r' - r)/r^2 \propto r_+ - r$. The contribution is $\propto (r_+ - r)^{3/2}$, and since $r_+ - r \propto J - J_*$, C has a singularity $(J - J_*)^{3/2}$: a tail $t^{-5/2}$ at frequency $\Omega(J_*)$, inside the band, for any F_{eq} . The same holds at pericentre.

None of this depends on a test function: the free part of $\delta\Phi$ decays algebraically for every distribution, the Gaussian included.

Tails of h_1 and of any test function. For h_1 the integrand is $B F_{\text{eq}} s$. If B behaves as J^p near $J = 0$ and does not vanish at J_t , the tail rule gives, with $F_{\text{eq}} \sim J^m$ and $s \sim J^{p_s}$:

- at $J = 0$ (Ω_{max}), a tail $t^{-(p+m+p_s+1)}$;
- at the vacuum edge, the same $t^{-(g+1)}$ as $\delta\Phi$;
- no turning-point tail if B is smooth in J , because h_1 does not look at a fixed radius.

The exponent p is not entirely free: a weight continuous at the circular orbit requires $p \geq 1/2$. The generic case is $p = 1/2$, and $\delta\Phi$ is generic ($C \propto \sqrt{J}$). The B of the code has $p = 2$, which speeds up the tail of the Ω_{max} edge by $3/2$ powers (Table 1).

What depends on B is therefore not the tail of $\delta\Phi$ but how much of it h_1 shows. The nearly circular orbits, which produce the tail at Ω_{max} , hardly weigh in h_1 with $p = 2$, and h_1 does not see the turning-point tail. $\delta\Phi$ is itself a projection on a “test function”, the Green's function, which nobody chose and which has kinks; h_1 is the projection on a smooth, chosen function. The dynamics is the same, and the collective mode of Section 1.7 does not depend on the observable either: the observable decides what is seen alongside it.

contribution	$\delta\Phi$, or h_1 with $p = \frac{1}{2}$	h_1 with $B \sim J^p$	code h_1 , $p = 2$
$J = 0, F_{\text{eq}}(0) > 0, s = 1$	$t^{-3/2}$	$t^{-(p+1)}$	t^{-3}
$J = 0, F_{\text{eq}}(0) > 0, s = \sqrt{J/J_t}$	t^{-2}	$t^{-(p+3/2)}$	$t^{-7/2}$
$J = 0, F_{\text{eq}}(0) > 0, s = (J/J_t)^{3/2}$	t^{-3}	$t^{-(p+5/2)}$	$t^{-9/2}$
$J = 0, F_{\text{eq}} \propto J^2, s = 1$	$t^{-7/2}$	$t^{-(p+3)}$	t^{-5}
vacuum edge, $F_{\text{eq}} \propto (J_t - J)^g$	$t^{-(g+1)}$	$t^{-(g+1)}$	$t^{-(g+1)}$
turning points (at fixed radius only)	$t^{-5/2}$	absent	absent
regenerated component, $F'_{\text{eq}}(0) \neq 0$	t^{-2}	$t^{-(p+3/2)}$	$t^{-7/2}$

Table 1: Exponents of the free tails for each observable. The first column holds for any generic observable, $\delta\Phi$ included; the others are those of h_1 with a test function that vanishes as J^p . The last row is the component regenerated by self-gravity (Section 1.7).

1.7 Self-gravity: frequency shift, collective mode and the η criterion

Self-gravity does two things. It changes the equilibrium: it deepens the potential and shifts the orbital frequencies. And it couples the perturbation to itself: δF produces a $\delta\Phi$ that acts back on the particles, and this coupling can sustain a collective mode. The η criterion uses the first effect as a measure of the second.

The frequency shift. Self-gravity deepens the potential and speeds up the orbits, so the whole band moves up. The nearly circular orbit, whose frequency is κ , shows by how much. Since $\Phi'(r_c) = L_0^2/r_c^3$, the definition of κ can be written $\kappa^2 = \Phi'' + 3L_0^2/r_c^4 = \Phi'' + 3\Phi'/r_c$. Poisson's equation in spherical symmetry, $\Phi'' + 2\Phi'/r = 4\pi G\rho$, eliminates Φ'' , and with $\Phi' = GM(< r)/r^2$,

$$\kappa^2 = 4\pi G \rho(r_c) + \frac{GM(< r_c)}{r_c^3}.$$

The radial frequency depends on two things: the enclosed mass, as for Kepler orbits, and *the local density at the radius of the orbit*. The second is the important one.

For the isochrone with $L_0 = 2$, $r_c = 5.742$. The density of the isochrone of unit mass and scale is $\rho = [3(1+a)a^2 - r^2(1+3a)]/[4\pi(1+a)^3a^3]$ with $a = \sqrt{1+r^2}$; at r_c it is 1.09×10^{-4} , and the enclosed mass is 0.696. Then $4\pi\rho = 1.37 \times 10^{-3}$, $M/r_c^3 = 3.68 \times 10^{-3}$ and $\kappa = 0.0711$, equal to $\Omega(0) = c^{-3} = 0.0711$: the epicyclic approximation is exact in the circular limit, as it must be.

The component of mass a_0 lives in a shell around r_c whose half-width is the typical epicycle radius. For the Gaussian, taking $J = \sigma_J$ (the maximum of $J^2 e^{-J^2/\sigma_J^2}$), $R_e = 0.92, 1.19$ and 1.68 for $\sigma_J = 0.03, 0.05$ and 0.10 . The maximum of $J^2(J_t - J)^2$ is at $J_t/2 = 1.4\sigma_J$, which gives a similar shell. The Wilson Maxwellian, instead, has its maximum at $J = 0$, and its mean action is $0.20 J_t = 0.027$ for $J_t = 0.138$: $R_e = 0.87$ against 1.19 , a narrower and denser shell.

Approximation 5: uniform shell. If the mass is spread uniformly over $[r_c - R_e, r_c + R_e]$, its density is $\rho_{\text{self}} \approx a_0/(4\pi r_c^2 2R_e)$, and half of the mass lies inside r_c . The formula above then gives

$$\delta\kappa^2 \approx \frac{a_0}{2R_e r_c^2} + \frac{a_0/2}{r_c^3}, \quad \frac{\delta\kappa}{\kappa} = \frac{\delta\kappa^2}{2\kappa^2}.$$

Section 3.4 compares this estimate with the measured shift.

The linearized equation. The Vlasov equation is $\partial_t F + \{F, H\} = 0$, with the Poisson bracket in angle-action variables $\{F, H\} = \partial_Q F \partial_J H - \partial_J F \partial_Q H$. With $F = F_{\text{eq}}(J) + \delta F$ and

$H = H_0(J) + \delta\Phi(Q, J, t)$, where H_0 is the Hamiltonian of the equilibrium and $\partial_J H_0 = \Omega$, the first-order terms are

$$\partial_t \delta F + \Omega(J) \partial_Q \delta F - F'_{\text{eq}}(J) \partial_Q \delta \Phi = 0.$$

The second term is the free transport of Section 1.6; the third is the field of the perturbation acting on the equilibrium. $\delta\Phi$ is in turn the potential of δF , through Poisson's equation, so the system is closed. `lineal.py` solves exactly this system (Section 2.3).

Modes. For a solution $\propto e^{i(kQ - \omega t)}$ the equation gives

$$\delta F_k(J) = \frac{k F'_{\text{eq}}(J)}{k \Omega(J) - \omega} \delta \Phi_k(J),$$

with $\delta\Phi_k(J)$ the coefficient k of the potential along orbit J . The denominator vanishes where $k\Omega(J) = \omega$: these are the *resonant* particles. δF_k produces a density and, through Poisson's equation, a $\delta\Phi$. Requiring it to be the one assumed gives a self-consistency condition whose solutions are the modes. Schematically,

$$\epsilon(\omega) = 1 - (\text{coupling} \propto G a_0) \int dJ \frac{k F'_{\text{eq}}(J) |\Psi(J)|^2}{k \Omega(J) - \omega} = 0,$$

with Ψ the radial shape of the mode. In the actual problem the coupling is a matrix in a basis of radial functions (Kalnajs's method), not a scalar; the scalar form suffices for the argument.

Where the mode can live. If ω lies inside the band, the integral has a real pole and the root moves into the complex plane with negative imaginary part: the mode is damped (Landau damping). If ω lies outside, there is no pole, the root is real and the mode oscillates without damping. We write the root as $\omega - i\gamma$, so that the mode behaves as $e^{-\gamma t} \cos(\omega t + \dots)$:

- ω is the *frequency* of the mode and $\gamma \geq 0$ its *damping rate*;
- $x = (\omega - \Omega_{\min})/\Delta\Omega$ is its *position in the band*: $0 < x < 1$ is inside (resonant, damped) and $x < 0$ below (a discrete mode, $\gamma = 0$).

The Antonov form. For radial perturbations of a spherical system there is a more precise form, due to Antonov: the linearized equation can be written $\partial_t^2 \psi = -\mathcal{A}\psi$ with $\mathcal{A}$ a self-adjoint operator. Two of its properties are used here.

- **Stability.** If $\partial F_{\text{eq}}/\partial E < 0$ over the whole support (the distribution is *monotonic*), $\mathcal{A}$ is positive and $\omega^2 > 0$: there are no growing radial modes. An oscillation that does not decay in a monotonic distribution is a genuine neutral mode, not an instability. In the work of Hadžić and collaborators the operator is defined in a space with weight $1/|\partial F/\partial E|$, which requires this monotonicity; this should be confirmed in the section on hypotheses of the reference paper.
- **Where the mode appears.** The essential spectrum of $\mathcal{A}$ is $\{k^2 \Omega(J)^2\}$, the band of Section 1.6. The gravitational term of $\mathcal{A}$ is *attractive* and tends to lower the eigenvalues, so a discrete mode, if it exists, appears *below* $\Omega_{\min}^2$, or in a gap between the bands of consecutive k . These gaps exist when $\Omega_{\max}/\Omega_{\min} < 2$; with the width of the Landau runs this ratio is 1.37.

The regenerated component at $\Omega_{\max}$. *Approximation 6.* The formula for δF_k shows what self-gravity generates near the circular orbit. There $\delta\Phi_1(J)$, the coefficient of the potential along the orbit, is proportional to $R_e \propto \sqrt{J}$, by the same argument that gives $C \propto \sqrt{J}$ (Section 1.6). Then $\delta F_1 \propto F'_{\text{eq}}(0) \sqrt{J}$ near $J = 0$: self-gravity generates a component with the regularity of $s = \sqrt{J/J_t}$, with an amplitude proportional to the coupling and *independent of the initial condition*. Its tail is t^{-2} at $\Omega_{\max}$ in any generic observable, $\delta\Phi$ included, and $t^{-7/2}$ in h_1 with $p = 2$ (Table 1).

In a monotonic distribution $F'_{\text{eq}}(0) \neq 0$ (for the Maxwellian, $F'_{\text{eq}} = -F_{\text{eq}} \Omega/T$ far from the edge), so the component is always present. In a distribution that vanishes as J^2 , $F'_{\text{eq}}(0) = 0$ and it does not appear. This is not a defect of the distribution but of the fixed- L model: with distributed L , $\kappa(L)$ varies and blurs the edge $\Omega_{\max}$. Section 3.8 confirms the component.

The η criterion. *Approximation 7: the shift as a measure of the coupling.* The shift of the mean frequency of the equilibrium *is not* the distance travelled by the mode: it is an effect of the same coupling on the background state. It is used as a measure of the strength of the coupling because both are of the same order: both come from terms $\sim 4\pi G \rho_{\text{self}}$, as the formula for κ^2 shows. The question of whether the coupling suffices to pull the mode out of the band then becomes a comparison between this shift and the width of the band.

The shift is measured by comparing two averages of the same quantity, the mass-weighted mean frequency of Section 1.6:

- $\bar{\Omega}$, computed in the self-consistent equilibrium, with the mass a_0 included in the potential;
- $\bar{\Omega}_{\text{iso}}$, the same average for the same distribution of orbits (the same shape $\tilde{F}$, with the same J_t , g and W_0), computed in the isochrone alone, as if the component had no mass ($a_0 \rightarrow 0$). It is the mean frequency that these orbits would have without self-gravity.

The difference between the two is the effect of self-gravity. The definition is

$$\delta\Omega = \frac{\bar{\Omega} - \bar{\Omega}_{\text{iso}}}{\bar{\Omega}_{\text{iso}}}, \quad \boxed{\eta = \frac{\delta\Omega}{\Delta\Omega/\bar{\Omega}}}$$

$\delta\Omega$ is the fractional increase of the mean frequency due to self-gravity, and $\Delta\Omega/\bar{\Omega}$ is the relative width of the band with mass: η is the relative shift divided by the relative width, a dimensionless number. This is how `eta.py` computes it, and all values of η in this document are defined in this way.

For example, for the Wilson model with $J_t = 0.138$ and $a_0 = 0.075$ (case A4 of Section 3), the mean frequency without self-gravity is $\bar{\Omega}_{\text{iso}} = 0.0687$, a mean radial period of 91.4. With the mass, $\bar{\Omega} = 0.0820$, so $\delta\Omega = 0.193$: self-gravity speeds up the orbits by 19%. The band with mass is $[0.0698, 0.0857]$, its relative width is $\Delta\Omega/\bar{\Omega} = 0.194$, and $\eta = 0.193/0.194 = 1.00$.

$\bar{\Omega}_{\text{iso}}$ is not $\Omega(0)$. $\Omega(0) = \kappa$ is the frequency of a single orbit, the circular one, the fastest in the band; $\bar{\Omega}$ and $\bar{\Omega}_{\text{iso}}$ are averages over all the orbits of the distribution.

η is a design criterion with a physical basis, not a theorem. What η guarantees is that the cases explore the range where the transition *can* occur, instead of lying entirely on one side of it. Section 3.7 shows that η orders the transition but does not fix it alone. The exact criterion is the following.

The loop gain at the edge. The density sees only the part of δF that is even in Q , because $r(-Q) = r(Q)$. With $b_k = \delta F_k + \delta F_{-k}$, the formula for δF_k above gives

$$b_k(J) = R_k(J; \omega) \delta\Phi_k(J), \quad R_k(J; \omega) = \frac{2k^2 \Omega(J) F'_{\text{eq}}(J)}{k^2 \Omega(J)^2 - \omega^2}, \quad k \geq 1,$$

and the potential of the shells closes the loop:

$$\delta\Phi_k(J) = \sum_{k' \geq 1} \int M_{kk'}(J, J') b_{k'}(J') dJ',$$

$$M_{kk'}(J, J') = 4\pi L_0 \iint \cos kQ \cos k'Q' G(r(Q, J), r(Q', J')) dQ dQ',$$

with $G(r, r') = -1/\max(r, r')$. This is the matrix form of the response method, with the harmonics in Q as the basis along the orbits. A mode of frequency ω exists if $MR(\omega)$, which maps a potential to the potential of the response it produces, has the eigenvalue 1. M is negative, since gravity attracts. With $F'_{\text{eq}} < 0$ and $\omega < \Omega_{\min}$ so is R , and MR is similar to the symmetric positive operator $|R|^{1/2}(-M)|R|^{1/2}$. Its largest eigenvalue $\lambda(\omega)$ is the gain of the loop. λ grows with ω up to the edge and decreases along imaginary frequencies, so a value $\lambda(0) \geq 1$ would imply a growing mode, and Antonov's stability gives $\lambda(0) < 1$. Hence there is a discrete mode below the band if and only if

$$\lambda_{\text{edge}} \equiv \lim_{\omega \rightarrow \Omega_{\min}^-} \lambda(\omega) > 1,$$

and its frequency is the root of $\lambda(\omega_d) = 1$. In quantum mechanics this is the Birman–Schwinger criterion for a bound state at the edge of a band. Near the edge $|R_1| \propto (J_t - J)^{g-1}/(\Omega - \omega)$, so λ_{edge} is finite only for $g > 1$, and

$$\lambda_{\text{edge}} - \lambda(\Omega_{\min} - \delta) \propto \delta^{\min(g-1, 1)},$$

with a logarithm for $g = 2$; for $g \leq 1$, λ diverges at the edge. Inverting this law gives the distance δ_d of the mode below the edge. Near the threshold, $\delta_d \propto (\lambda_{\text{edge}} - 1)^{1/(g-1)}$ for $1 < g < 2$, and $\delta_d \propto \lambda_{\text{edge}} - 1$ up to the logarithm for $g \geq 2$. For $g < 1$ there is a mode at any coupling, at a distance that decreases as the power $1/(1 - g)$ of the coupling, and for $g = 1$ the distance decreases exponentially. The threshold $\lambda_{\text{edge}} = 1$ does not depend on the family; Section 3.9 computes it for all the equilibria.

1.8 The vacuum edge and the Hadžić dichotomy

Hadžić, Rein, Schrecker and Straub (*Damping versus oscillations for a gravitational Vlasov–Poisson system*, Arch. Ration. Mech. Anal. **249**, 45, 2025; [arXiv:2301.07662](#)) work in *the same reduction as this document*: the radial Vlasov–Poisson system with all particles at the same $|L|$, in the fixed potential of a central point mass M , with small equilibria $f = \varepsilon (E_0 - E)_+^k$ (ε is a size parameter, the analogue of our a_0). Their Theorem 1.2 states the following.

- If $1/2 < k \leq 1$, there is $\varepsilon_0(k) > 0$ such that for $0 < \varepsilon < \varepsilon_0$ the linearized problem is *not* damped: there is at least one eigenvalue below the bottom of the essential spectrum, in the principal gap, that is, a discrete mode with frequency below $\Omega_{\min}$.
- If $k > 1$, there is $\varepsilon_0(k) > 0$ such that for $0 < \varepsilon < \varepsilon_0$ there is Landau damping (in a weak sense, without a rate) and the point spectrum is empty.

Only the regularity at the vacuum edge matters: the result holds for any F that behaves as $(E_0 - E)^k$ there, and the authors note that the King model ($k = 1$) is not damped. The proof has two parts: the absence of eigenvalues embedded in the essential spectrum, and a Birman–Schwinger principle for the eigenvalues in the principal gap.

The two differences from the present setting are the background, an isochrone instead of a point mass (both give a radial period that is monotonic in the energy, which is the property the proof uses), and the size: the theorem holds for $\varepsilon < \varepsilon_0(k)$ and does not estimate ε_0 . For $k = g > 1$ nothing prevents a discrete mode from appearing at finite mass. The linear calculations of

Section 3 estimate that threshold for $g = 2$ and test the case $g \leq 1$ at small mass. Related work: Ramming and Rein (Phys. D **365**, 72, 2018) study numerically oscillating solutions of the radial problem without a central mass; Rioseco and Sarbach (Class. Quantum Grav. **37**, 195027, 2020) study phase mixing in an external central potential; and Hadžić, Rein and Straub (Arch. Ration. Mech. Anal. **243**, 611, 2022) prove the existence of linearly oscillating galaxies.

The tail rule gives an intuition for why the edge matters. An edge at which the weight vanishes as $(\Omega - \Omega_{\min})^g$ produces in the free response a tail $\propto t^{-(g+1)}$, slower for smaller g . With self-gravity, a free response that lingers near the edge gives the coupling more time to feed it back and sustain an oscillation. This is a heuristic, not the proof.

The edges of the three families are the following.

- **The Gaussian** has no edge at $\Omega_{\min}$: it switches off smoothly.
- **The hollow family** has a vacuum edge at J_t with exponent $g = 2$, and vanishes as J^2 at the circular orbit.
- **The Wilson Maxwellian** has a genuine vacuum edge at J_t , the low-frequency end of the band towards which the coupling pushes the mode: $F \propto (E_t - E)^2$, with $g = 2$, on the damped side and away from the marginal case $g = 1$, which is King. At $J = 0$ it is maximal, as is every monotonic distribution and as are the equilibria of that work.

With the Maxwellian the edge mechanism is not switched off but fixed in a known class, and the edge experiment consists in changing a single number, g , at fixed η . Since it is a linear question, it is answered with the linear solver (Section 3.8).

1.9 Nonlinear scales and the limits of linear theory

The perturbative objection. An initial condition $F = F_{\text{eq}} [1 + \varepsilon s \chi(Q)]$ suggests a perturbation, and with it a predictable objection: if the problem is linear, linear theory (Kalnajs's response, or the linearized solver of this repository) solves it, and the particle simulation adds nothing. Part of the objection is correct, and the design accepts it. Every question that is linear (whether a discrete mode exists, the values of γ and ω , and how they depend on η and on the edge) is answered here with `lineal.py`, without particles. This is cheap and precise, and it is the natural comparison with the theorems of Hadžić and collaborators, which are linear (Section 3). The PIC runs in the linear regime serve as validation, not as results. But the conclusion that the code is unnecessary does not follow, for the reasons below.

The parameter is not ε . In a fixed potential the Vlasov equation is *exactly linear* in F : the free solution of Section 1.6 holds for any ε . Nonlinearity enters only through $\delta\Phi$, which is proportional to the mass, to the amplitude and to how much potential the shape of the perturbation produces, $\delta\Phi \propto a_0 \varepsilon \rho_\Phi$, with ρ_Φ defined in Section 3.3. With $\varepsilon = 1$, $\delta F/F$ is of order one, and the problem can still be linear if the mass is small; conversely, a small ε can make it nonlinear, for the reasons that follow. That the initial condition looks like a perturbation says nothing: what decides is the size of $\delta\Phi$ compared with two rates.

Two nonlinear scales. *Approximation 8: the pendulum.* Near a resonance $k\Omega(J_r) = \omega$, a particle feels the Hamiltonian $H = H_0(J) + |\delta\Phi_k| \cos(kQ - \omega t)$. In the frame rotating with the wave, with the angle $\psi = kQ - \omega t$ and its conjugate momentum $I = J/k$, the Hamiltonian is $K = H_0(kI) - \omega I + |\delta\Phi_k| \cos\psi$. Expanding about the resonance, $H_0(J) \approx H_0(J_r) + \Omega(J_r)(J - J_r) + \frac{1}{2}\Omega'(J_r)(J - J_r)^2$, the linear term cancels with $-\omega I$, and

$$K \approx \frac{1}{2} k^2 \Omega'(J_r) (I - I_r)^2 + |\delta\Phi_k| \cos\psi,$$

a pendulum. Its small oscillations have the *bounce frequency*, and its separatrix sets the half-width of the *trapping island*:

$$\omega_b = k \sqrt{|\Omega'(J_r)| |\delta\Phi_k|}, \quad \Delta J_{\text{atr}} = 2\sqrt{|\delta\Phi_k|/|\Omega'(J_r)|}.$$

Particles inside the island are trapped by the wave and oscillate about its phase. The approximation assumes an isolated resonance and an amplitude that changes little over one bounce period.

ω_b is compared with two rates, and each comparison is a different nonlinear scale.

- $\mu = \omega_b/\Delta\Omega$: trapping versus phase mixing of the *continuum*. If $\mu \ll 1$ the particles dephase before they feel the well; if $\mu \gtrsim 1$ islands form across the whole band.
- $\nu = \omega_b/\gamma$: trapping versus damping of the *mode*. It is O’Neil’s parameter (1965, Phys. Fluids 8, 2255) for Landau damping in plasmas. If $\nu \ll 1$ the wave is damped before it traps, and linear theory holds at all times. If $\nu \gtrsim 1$, from $t \sim 2\pi/\omega_b$ on, the resonant particles are trapped, damping stops, and the amplitude oscillates at a frequency $\sim \omega_b$ and saturates. The final state, with a plateau of F around J_r , is not predicted by linear theory, which holds only for $t \lesssim 1/\omega_b$.

Approximation 9: the amplitude of $\delta\Phi$. The naive estimate $|\delta\Phi| \sim \varepsilon |\Phi_{\text{self}}|$ is wrong by a large factor. A $\cos Q$ perturbation bunches the particles in phase: it does not change the mass of the shell, it only *displaces* it radially by a fraction $\sim \varepsilon R_e$ of its radius, and the potential of a displaced shell differs from the original one by $\sim \varepsilon (R_e/r_c) |\Phi_{\text{self}}|$. Section 3.10 confirms this with a reference run. In practice no estimate is needed: the linear solution of each case gives $\delta\Phi(r, t)$ with $\varepsilon = 1$, and

$$\omega_b \approx \sqrt{|\Omega'| \max_r |\delta\Phi(r, 0)|}, \quad |\Omega'| \approx \Delta\Omega/J_t,$$

with $k = 1$ and the mean slope of $\Omega(J)$ instead of its value at the resonance. For another ε , ω_b scales as $\sqrt{\varepsilon}$.

The transition is nonlinear. *Approximation 10: O’Neil’s criterion.* It is a result from plasma physics (one wave, one dimension), and here it is used as a scale, not as a quantitative prediction: it is what the PIC runs test.

As $\eta \rightarrow \eta_c$ the damping rate $\gamma \rightarrow 0$, while ω_b , which is set by the amplitude, stays finite. So $\nu \rightarrow \infty$ for any finite amplitude: **linear theory predicts where the transition is, but near it does not describe the late-time evolution for any finite amplitude.** This is precisely the regime to be studied, and it is where the code is indispensable. On the other side ($\eta > \eta_c$) the mode is discrete and resonates with no particle, so O’Neil’s mechanism does not act, and the nonlinear question is a different one: whether the mode survives at finite amplitude. Section 3.10 gives the values of ν for the cases considered.

State of the theory. The results of Hadžić, Rein, Schrecker and Straub concern the *linearized* problem at small mass ($\varepsilon < \varepsilon_0$): they state when $\delta\Phi$ decays and when there are pure oscillations, for the linear dynamics around the equilibrium. The closest nonlinear numerical work they cite is that of Ramming and Rein (2018), for the radial problem without a central mass. To our knowledge, whether this damping persists in the nonlinear gravitational problem is an open question. In plasmas there is the theorem of Mouhot and Villani (2011), for small and very regular perturbations; in gravity we know of no equivalent, and O’Neil’s argument shows why the question is delicate precisely near the transition. This is the question that the code can address and linear theory cannot.

Consequences for the design.

1. The linear questions are answered with `lineal.py`: the linear calculations of Section 3 are done, not proposed.
2. The PIC runs concentrate where $\nu \gtrsim 1$, with the linear solution of the same equilibrium as reference at each point. What is measured is the *deviation* from it, as a function of ν and of time.
3. The perturbative form of the initial condition becomes an advantage. As a sentence for the paper: *we use $F = F_{\text{eq}}(1 + \varepsilon s\chi)$ not because we assume linearity, but because it is a controlled family in which the linear prediction is computed exactly, so that the nonlinear deviation is measured without ambiguity.*

2 Numerical methods

2.1 Self-consistent equilibria

`equilibrio.py` constructs the equilibrium by iterating

$$\Phi_{\text{self}} \rightarrow \text{map}(r, p_r) \rightarrow (Q, J) \rightarrow F_{\text{eq}}(J) \rightarrow \rho(r) \rightarrow \text{Poisson} \rightarrow \Phi_{\text{self}},$$

starting from $\Phi_{\text{self}} = 0$, until the potential changes by less than 10^{-13} between iterations. The angle-action map of each iterate is computed numerically (`aa_numerico.py`): for each energy the turning points $r_{\pm}$ are found by bisection, and the integrals are computed with Gauss-Legendre quadrature after the substitution $r = r_m + r_a \sin \theta$, where r_m and r_a are the centre and half-width of $[r_-, r_+]$. At a turning point $E - \Phi_{\text{ef}}$ vanishes linearly in r , so $p_r \propto |r - r_{\pm}|^{1/2}$: the integrand of J has an infinite derivative there, and that of the radial period, $1/p_r$, is infinite. Gauss-Legendre quadrature converges slowly for both. With the substitution, $E - \Phi_{\text{ef}} = r_a^2 \cos^2 \theta h(\theta)$ with h smooth and positive, so $p_r dr/d\theta = r_a^2 \cos^2 \theta \sqrt{2h}$ and $(dr/d\theta)/p_r = 1/\sqrt{2h}$ are smooth, and 64 nodes give J to 10^{-14} in the bare isochrone (Section 2.1 of the demo document). $E(J)$ is tabulated, and $\Omega = dE/dJ$ is obtained by finite differences.

Convergence. This is a fixed-point (Picard) iteration without relaxation. If the map is a contraction near the solution, with factor $q < 1$, the iteration converges geometrically, the fixed point is unique in that neighbourhood, and the error after the last iteration is at most $q/(1 - q)$ times the last change. For small mass the contraction holds, with $q = O(a_0)$: Φ_{self} is the potential of a fixed mass, and a change of the potential only redistributes that mass. For finite mass, `equilibrio.py` measures q as the median ratio of consecutive changes and reports the bound. For the Wilson and King equilibria used in the PIC runs, q grows from 0.017 at $a_0 = 0.0065$ to 0.587 at $a_0 = 0.5$, following $q \approx a_0/(a_0 + 0.35)$ for $g = 2$, and the bound is below 10^{-13} . The script stops with an error if the change is still above the tolerance after 60 iterations, which by the fit would happen near $a_0 \approx 0.7$.

An independent construction. For the lowered Maxwellian, which depends on the energy only, the density at r depends only on the potential at r : $4\pi r^2 \rho = 8\pi^2 L_0 A \sqrt{2\pi T} E_\gamma (g + \frac{1}{2}, (E_t - \Phi_{\text{ef}})/T)$. Poisson's equation is then an ordinary differential equation, integrated over the support, and three numbers (E_t , T and A) are fixed by the mass, W_0 and J_t (`equilibrio_edo.py`; `equilibrio.py --metodo edo`). The two constructions agree to $1\text{--}2 \times 10^{-6}$ in Φ_{self} . The difference is the discretization error of the iteration, which decreases as the square of its radial spacing. Evolved with the PIC code at $\varepsilon = 0$, the equilibria of the two constructions agree to better than 1%. Their departures from stationarity come from the PIC discretization. There are two effects: an $O(\Delta r^2)$ error of the field, and a slow spread of the actions caused by the

discreteness of the particles, which disappears with four times more particles. The derivation and the tables are in Section 2.1 of the demo document (`docs/demo_eta/demo_eta.pdf`), and the PIC tests in its Section 5.2.

The initial data use nodes in $[0, J_{\max}]$ (Section 2.2). With compact support $J_{\max} = J_t$, the edge of the support; for the Gaussian, $J_{\max} = 6\sigma_J$.

2.2 Particle loading

The particles are placed on a regular grid of the true angle-action variables of the equilibrium:

$$J_i = (i - \frac{1}{2}) \frac{J_{\max}}{N_{rc}}, \quad i = 1 \dots N_{rc}; \quad Q_j = (j - \frac{1}{2}) \frac{2\pi}{N_{pc}}, \quad j = 1 \dots N_{pc}.$$

Each node is a particle of weight $8\pi^2 L_0 F(Q_j, J_i) \Delta Q \Delta J$, and its position (r, p_r) is obtained by inverting the map of the equilibrium. There are $N = N_{rc} N_{pc}$ particles. This is a quiet start: there is no random sampling noise, and the remaining noise comes from the discretization. The same grid in J sets the recurrence ceiling (Section 4.2).

2.3 Linearized solver

The linear calculations have two steps. `lineal.py` computes the time evolution of the problem linearized about each equilibrium, without particles, and `barrido_lineal.py` extracts from the signal the collective mode: its frequency, its damping rate and its position in the band (Section 2.4).

The equations. The problem is the linearized Vlasov–Poisson system of Section 1.7, written in the angle-action variables of the equilibrium:

$$\partial_t \delta F + \Omega(J) \partial_Q \delta F = F'_{\text{eq}}(J) \partial_Q \delta \Phi(Q, J, t), \quad (1)$$

$$\delta \Phi(Q, J, t) = \delta \Phi(r(Q, J), t), \quad \delta \Phi(r, t) = 8\pi^2 L_0 \iint dQ' dJ' G(r, r(Q', J')) \delta F(Q', J', t). \quad (2)$$

The first equation is the evolution of the perturbation: free transport on the left, and the push of the self-consistent field on the equilibrium on the right. The second is Poisson’s equation: the potential of the perturbation is the sum of the contributions of all points of phase space, with the Green’s function of shells $G(r, r') = -1/\max(r, r')$ (Section 1.5). $r(Q, J)$ is the radius of the point (Q, J) in the equilibrium. In linear theory this map is fixed: the changes of the orbits produced by $\delta \Phi$ are of second order, and they enter only through the term on the right.

Initial data. $\delta F(Q, J, 0) = F_{\text{eq}}(J) s(J) \cos Q$, the perturbation of Section 1.4 with $\varepsilon = 1$. Since the problem is linear, the response to another ε is ε times this one, and all reported quantities (h_1 , $\delta \Phi$, and ω_b with $\varepsilon = 1$) scale accordingly.

Time evolution rather than eigenvalues. A Landau-damped mode *is not* an eigenvector of the linear operator: the spectrum on the real axis is the continuous band plus, if they exist, the discrete modes below it. The damped mode is a pole of the analytic continuation of the response, and computing it as an eigenvalue (the matrix method of Kalnajs) requires that continuation. In the time evolution it appears by itself, as a term $e^{-\gamma t}$ in the late signal, together with the algebraic tails of the continuum (Section 1.6); a discrete mode appears as an oscillation of constant amplitude. Solving the evolution and fitting the tail gives both without analytic continuation, and it is directly comparable with the PIC run.

Discretization. The solver uses the same regular grid as the PIC initial data (Section 2.2), refined:

$$J_i = \left(i - \frac{1}{2}\right) \frac{J_{\max}}{N_J}, \quad Q_j = \left(j - \frac{1}{2}\right) \frac{2\pi}{N_Q}, \quad N_J \times N_Q = 1600 \times 32.$$

The unknown is $\delta F_{ij}(t) = \delta F(Q_j, J_i, t)$. What the method needs from the equilibrium is computed once:

- **the radii** $r_{ij} = r(Q_j, J_i)$, by inverting the angle-action map of the equilibrium (`equilibrio.invertir`): 51 200 inversions, about 16 s, stored in a file;
- **the frequencies** $\Omega_i = \Omega(J_i) = dE/dJ$, from the table $E(J)$ of the equilibrium by centred differences, which reproduce $\Omega = (J + c)^{-3}$ of the isochrone to 10^{-4} ;
- **the derivative** $F'_{\text{eq}}(J_i)$, analytic: that of the formula for the Gaussian and the hollow family, and $F'_{\text{eq}} = (dF_{\text{eq}}/dE) \Omega(J)$ for the Maxwellian, which is written in the energy.

Transport. Without the term on the right, the solution is $\delta F(Q, J, t + \Delta t) = \delta F(Q - \Omega(J)\Delta t, J, t)$: each row of fixed J rotates in Q without deforming. In Fourier space this is a phase: with $\widehat{\delta F}_{ik} = \frac{1}{N_Q} \sum_j \delta F_{ij} e^{-ikQ_j}$, the fast Fourier transform along Q at each J_i ,

$$\widehat{\delta F}_{ik}(t + \Delta t) = e^{-ik\Omega_i\Delta t} \widehat{\delta F}_{ik}(t).$$

It is exact for any Δt : there is no stability condition and no numerical dispersion. The only errors come from the finite number of harmonics, $|k| < N_Q/2$, and the finite number of rows in J , which sets the recurrence ceiling $T_{\text{rec}}(k) = N_J\tau_1/k$ (Section 4.2): with $N_J = 1600$ it is $1600\tau_1$ for $k = 1$, far beyond the runs.

Field kick. Without the transport, the equation is $\partial_t \delta F = F'_{\text{eq}}(J) \partial_Q \delta \Phi$. Over a short interval h , with $\delta \Phi$ fixed,

$$\delta F_{ij} \leftarrow \delta F_{ij} + h F'_{\text{eq}}(J_i) \partial_Q \delta \Phi(Q_j, J_i),$$

with the derivative in Q also computed in Fourier space: transform along Q , multiply by ik and transform back.

Poisson's equation at the nodes. Each node represents a spherical shell of mass $m_{ij} = 8\pi^2 L_0 \delta F_{ij} \Delta Q \Delta J$ at radius r_{ij} : the set of particles at that radius with all possible orientations of the angular momentum. The potential of a set of shells is exact with the Green's function of shells: at the radius of node l ,

$$\delta \Phi(r_l) = -\frac{1}{r_l} \sum_{r_n < r_l} m_n - \sum_{r_n > r_l} \frac{m_n}{r_n} - \frac{m_l}{r_l},$$

where the first term is the enclosed mass, the second the exterior shells and the third the node's own shell. G is continuous at $r' = r$ (it is $-1/r$ on both sides; its derivative has the kink), so the contribution of the own shell is unambiguous. The code splits it half into each sum, which gives the same result and treats nodes with equal radii symmetrically. The masses can be negative, because δF is a perturbation. Since the radii do not change, the ordering of the nodes by radius is computed once, and each evaluation of $\delta \Phi$ consists of two cumulative sums over the $N_J N_Q$ nodes. This $\delta \Phi$ at each node is the one used in the kick; for the output it is also interpolated at 121 radii in $r \in [3, 15]$.

The full step. The two operations are combined by Strang splitting: half a kick, a full transport step and another half kick,

$$\delta F(t + \Delta t) = \mathcal{K}_{\Delta t/2} \mathcal{T}_{\Delta t} \mathcal{K}_{\Delta t/2} \delta F(t),$$

which is second order in Δt . The error per step comes from the fact that transport and kick do not commute, and it is of the order of the square of the product of Δt and the scale of the coupling. With $\Delta t = 0.5$ it is negligible, as checked by halving Δt . With the option `--libre` the kicks are omitted, which gives the exact free phase mixing of Section 1.6. A case of $10 \tau_1$ takes between about 2 000 and 22 000 steps, depending on τ_1 , and costs one to three minutes.

Recorded quantities. Every two time units the solver stores:

- $h_1(t) = \sum_{ij} \delta F_{ij} B(J_i) e^{-iQ_j} / (N_Q \sum_i F_{\text{eq}}(J_i) B(J_i))$, with the test function of Section 1.5, and h_2 with e^{-2iQ_j} . With the flat perturbation $h_1(0) = \frac{1}{2}$. This is exactly the analogue of $[h_1(\varepsilon) - h_1(0)]/\varepsilon$ in the PIC runs: the same frame, the same B and the same normalization, which is why the two agree to four digits in `11_landau`;
- $\delta\Phi(r, t)$ at the 121 radii, with the amplitude of $\varepsilon = 1$.

2.4 Extraction of the mode

Approximation 11: the dominant pole. The mode is extracted from the late signal with the *matrix pencil* method, the same fit used for `11_landau`. We take N samples of h_1 in a window, $y_n = h_1(t_0 + n\Delta)$, and represent them as a sum of M exponentials,

$$y_n \approx \sum_{m=1}^M a_m z_m^n, \quad z_m = e^{s_m \Delta}.$$

The samples form the Hankel matrix $Y_{il} = y_{i+l}$, with $i = 0 \dots N-L-1$ and $l = 0 \dots L$, $L = N/2$. If the model were exact, each row would be a combination of the M vectors $(1, z_m, z_m^2, \dots, z_m^L)$, and Y would have rank M . The singular value decomposition $Y = U \Sigma V^H$ gives that subspace: the first M columns of V . Calling V_1 these columns without their last row and V_2 without their first row, a shift by one place amounts to a multiplication by the z_m , and the z_m are the eigenvalues of $V_1^+ V_2$ (V_1^+ is the pseudoinverse). Then $s_m = \ln z_m / \Delta$, and the amplitudes a_m follow by least squares. The mode is the term of largest amplitude, with

$$\omega = -\text{Im } s, \quad \gamma = -\text{Re } s,$$

because the signal behaves as $e^{-i\omega t - \gamma t}$. We use $M = 3$: one term for the mode and two to absorb what remains of the continuum.

Windows. The fit is repeated in windows $[a, 2a] \tau_1$ with $a = 1, 2, 3, 5, 10, 20$ and 30 , those covered by the run, subsampled to at most 1000 points (the Hankel matrix grows as N^2 ; with $\omega \approx 0.08$ the resulting sampling remains well below the Nyquist limit). Before $t = \tau_1$ the transient of phase mixing dominates, and it is not a pole. Two consecutive windows *agree* if their discrepancy,

$$\max \left(\frac{|\omega_1 - \omega_2|}{\Delta\Omega}, \frac{|\gamma_1 - \gamma_2|}{\max(\bar{\gamma}, 2 \times 10^{-3} \Delta\Omega)} \right),$$

is below 0.2; the floor $2 \times 10^{-3} \Delta\Omega$ avoids dividing by $\gamma \approx 0$ for discrete modes. We report the latest pair that agrees, because it is the asymptotic behaviour, and the value is the mean of the pair. If no pair agrees, the tail is not a single exponential (algebraic phase mixing, or two comparable components), and the case is marked “no dominant pole”. With ω and γ we give $x = (\omega - \Omega_{\text{min}}) / \Delta\Omega$ and $\gamma / \Delta\Omega$, with the band of the equilibrium.

Bounce frequency. The linear $\delta\Phi(r, t)$ gives $\omega_b = \sqrt{|\Omega'| \max_r |\delta\Phi|}$ with $|\Omega'| \approx \Delta\Omega/J_t$ (Approximation 9, Section 1.9), at $t = 0$ and at the start of the fitting window. With γ it gives $\nu = \omega_b/\gamma$ for any ε , since $\omega_b \propto \sqrt{\varepsilon}$.

Fit of $\delta\Phi$ as a control. The same procedure is applied to $\delta\Phi(r, t)$, which uses no test function. In each window we take the dominant time series of the singular value decomposition of the matrix $\delta\Phi(r_n, t_m)$ and apply matrix pencil with $M = 4$; since the signal is real, the poles come in conjugate pairs, and we take the one with $\omega > 0$ of largest amplitude.

2.5 Free-decay quadrature

Approximation 12: tails in the bare isochrone. The free decay of Section 3.3 is computed with `colas_libres.py`, which evaluates the integrals of Section 1.6 by quadrature, without time evolution, in the isochrone without self-gravity ($a_0 \rightarrow 0$): the tail is a property of the shape of F_{eq} . The quadrature reproduces to two digits the decay of `landau_libre` (the sum over the particles of `11_landau`) over $t = 400$ – 1400 , and changes by less than 3×10^{-4} with half the grid in J up to $40 \tau_1$.

2.6 Reference runs

The runs of Section 11 of the notes (`11_landau`) serve as a validated reference: a Gaussian with $\sigma_J = 0.10$ and $J_{\text{max}} = 0.6$, $a_0 = 0.01$, $L_0 = 2$, a perturbation $\varepsilon = 0.1$ with $\chi = \cos Q$, $N_{rc} \times N_{pc} = 400 \times 25$ ($N = 10^4$), a grid $r \in [0, 20]$ with $\Delta r = 0.1$, $\Delta t = 0.1$ (`courant` = 2), the yoshida4 integrator and the test function $j_1 = s_{j1} = 0.10$. Their η is 0.05, deep in the damped regime. The run `landau_libre` evolves the same perturbation without self-gravity, in the potential of the equilibrium.

3 Results of the linear theory

Each case of Table 2 is solved and fitted as explained in Section 2; we call this set of linear calculations block L. The solutions are in `exe/eta_lineal/` (outside version control), and rebuilding the tables from them takes seconds.

3.1 The cases

Each equilibrium has a name, the one used by `barrido_lineal.py`. Unless stated otherwise, the family is the Wilson Maxwellian ($g = 2$, $W_0 = 3$), with $J_t = 0.138$ and $L_0 = 2$.

3.2 Frequency bands

Table 3 gives the bands of the isochrone without self-gravity and compares the relative width with Approximation 3.

3.3 Free phase mixing

The free decay of the reference run. If the free decay of h_1 in `landau_libre` is Gaussian, $\ln(|h_1(0)|/|h_1(t)|) = \sigma_\Omega^2 t^2/2$ with constant σ_Ω . The data give Table 4.

A practical consequence: the free part of h_1 reaches the numerical floor near $t \approx 5 \tau_1$. Hence $t_{\text{fin}} = 10 \tau_1$ for the linear runs: half of the time for the free mixing to end and the other half to fit what remains. For $\delta\Phi$ the count is different.

case	equilibrium	a_0	η	role
L1, A1, L2, A2	reference	0.0030, 0.0065, 0.013, 0.020	0.05–0.30	scan in η
L3, A3, L4, A4	reference	0.031, 0.042, 0.058, 0.075	0.45–1.00	scan in η
L5, A5, L6	reference	0.097, 0.121, 0.170	1.24–2.00	scan in η
A6, A7	$J_t = 0.276$	0.044, 0.148	0.30, 1.00	degeneracy: width
A8, A9	$J_t = 0.083$	0.0109, 0.044	0.30, 1.00	degeneracy: width
D1, D2	$L_0 = 1, J_t = 0.094$	0.018, 0.069	0.29, 1.00	degeneracy: L_0
E1	hollow, $J^2(J_t - J)^2$	0.075	1.00	family
E2	Gaussian, $\sigma_J = 0.05$	0.069	0.97	family; bridge to 11_landau
E3	Wilson, $W_0 = 6$	0.073	1.00	family: thermal limit
H9, H3	hollow, $J_t = 0.083$	0.041, 0.100	0.99, 3.1	family
G075, G1, G15, G3	Wilson with $g = 0.75$, 1, 1.5, 3	0.0775–0.0739	1.00	edge exponent
G075a, G075b	Wilson with $g = 0.75$	0.0070, 0.0215	0.10, 0.30	sharp edge, weak coupling
G1a, G1b	Wilson with $g = 1$ (King)	0.0069, 0.0212	0.10, 0.30	sharp edge, weak coupling
landau	Gaussian, $\sigma_J = 0.10$	0.010	0.05	validation: 11_landau

Table 2: The equilibria of this document. Names with L are intermediate points of the scan; those with A, D and E are design cases; the G cases vary the exponent of the edge (with $g = 2$ the case is A4). The PIC runs (Section 4.3) have their own numbering: blocks N, V, Z and B.

support	width	support in J	band $[\Omega_{\min}, \Omega_{\max}]$	exact $\Delta\Omega/\Omega$	$3\Delta J/(J + c)$
Gaussian	$\sigma_J = 0.10$	[0.0061, 0.2763]	[0.0513, 0.0705]	0.315	0.317
	$\sigma_J = 0.05$	[0.0031, 0.1382]	[0.0601, 0.0708]	0.163	0.163
	$\sigma_J = 0.03$	[0.0019, 0.0829]	[0.0642, 0.0709]	0.099	0.099
compact	$J_t = 0.276$	[0, 0.276]	[0.0514, 0.0711]	0.322	0.324
	$J_t = 0.138$	[0, 0.138]	[0.0602, 0.0711]	0.166	0.167
	$J_t = 0.083$	[0, 0.083]	[0.0642, 0.0711]	0.101	0.101

Table 3: Bands of the isochrone with $L_0 = 2$ ($c = 2.414$), without self-gravity; in $\Delta\Omega/\Omega$, Ω is the centre of the band and J that of the interval. The linear approximation is accurate to better than 1 %, and the error grows with the width, as it should. Without self-gravity the band of compact support is the same for all families, and it extends up to $\Omega(0) = c^{-3}$ because no threshold cuts off the nearly circular orbits.

t	$ h_1 / h_1(0) $	inferred σ_Ω	$\Delta\Omega/\sigma_\Omega$
400	5.5×10^{-1}	2.74×10^{-3}	6.97
800	8.0×10^{-2}	2.81×10^{-3}	6.80
1000	1.6×10^{-2}	2.88×10^{-3}	6.63
1200	1.5×10^{-3}	3.00×10^{-3}	6.36
1400	9.9×10^{-5}	3.07×10^{-3}	6.23

Table 4: Free decay of 11_landau. σ_Ω varies by only 12 % while $|h_1|$ falls by four orders of magnitude: the decay is almost exactly Gaussian, as predicted by the transform of a smooth w . The slight late increase is the non-Gaussian part of w . The ratio $\Delta\Omega/\sigma_\Omega \approx 6.2$ –7.0 agrees with the expected 6.1 (Approximation 4).

Tails of the three families. Table 5 gives the exact free decay (Section 2.5). Its last column is

$$\rho_\Phi = \frac{[\|\delta\Phi(0)\|/(a_0 \varepsilon)]_{\text{family}}}{[\|\delta\Phi(0)\|/(a_0 \varepsilon)]_{\text{Gaussian}}},$$

the initial $\delta\Phi$ per unit mass and unit amplitude, relative to the Gaussian with $\sigma_J = 0.05$ and flat perturbation. It is used to compare amplitudes between families (Section 1.9).

F_{eq}	$s(J)$	$ h_1 / h_1(0) $ $10\tau_1$	$2\tau_1$	$\ \delta\Phi\ /\ \delta\Phi(0)\ $ $5\tau_1$	$10\tau_1$	slope $20-40\tau_1$	ρ_Φ
Gaussian	1	1.7×10^{-6}	7.9×10^{-2}	1.2×10^{-3}	1.2×10^{-4}	-2.4	1.00
$J^2(J_t - J)^2$	1	9.7×10^{-5}	4.3×10^{-2}	3.1×10^{-3}	4.2×10^{-4}	-3.0	1.09
Wilson, $W_0 = 3$	1	1.7×10^{-4}	1.9×10^{-1}	5.0×10^{-2}	1.8×10^{-2}	-1.5	0.66
Wilson, $W_0 = 3$	$\sqrt{J/J_t}$	5.6×10^{-5}	1.2×10^{-1}	2.0×10^{-2}	4.9×10^{-3}	-2.0	0.32
Wilson, $W_0 = 3$	$(J/J_t)^{3/2}$	4.3×10^{-5}	4.4×10^{-2}	2.4×10^{-3}	3.0×10^{-4}	-2.8	0.11
Wilson, $W_0 = 6$	$(J/J_t)^{3/2}$	3.6×10^{-5}	1.0×10^{-1}	7.7×10^{-3}	1.0×10^{-3}	-2.9	0.06

Table 5: Exact free decay with the reference width ($\sigma_J = 0.05$ or $J_t = 0.138$), bare isochrone with $L_0 = 2$. The $\delta\Phi$ columns use no test function; the h_1 column uses the one of the code, $B = J^2 e^{-(J-0.05)^2/0.05^2}$. “Wilson” is the lowered Maxwellian with $g = 2$. The slope is $d \ln \|\delta\Phi\| / d \ln t$ between 20 and $40\tau_1$. It confirms the exponents of Table 1, $-3/2$ and -2 , and with $s = (J/J_t)^{3/2}$ gives a tail from the edge $J = 0$ that is faster than the turning-point tail ($-5/2$).

Three conclusions follow.

1. A distribution with $F_{\text{eq}}(0) > 0$ and perturbation $s = 1$ leaves in $\delta\Phi$ a tail 150 times larger than the Gaussian at $10\tau_1$, at the frequency Ω_{max} . With $s = (J/J_t)^{3/2}$ it drops to the level of the Gaussian. This is why the Maxwellian uses `suave3`.
2. That smooth perturbation produces little potential, $\rho_\Phi = 0.11$: the mass of a thermal distribution is in nearly circular orbits, with small epicycles, where every smooth perturbation must vanish. This has to be compensated with a larger ε (Section 4.3).
3. For every family the free part of $\delta\Phi$ is still $\sim 10^{-4}$ of its initial value at $10\tau_1$, the order of the noise floor. The damping rate is therefore fitted against this free reference, without assuming that mixing is over.

3.4 The frequency shift

Table 6 compares the uniform-shell estimate (Approximation 5) with the shift measured by `eta.py`.

Four conclusions follow, all used in the design.

1. **The local-density term dominates:** with $\sigma_J = 0.05$ it contributes 83 % of $\delta\kappa^2$. This is why the shift is more than twice the naive estimate based on the enclosed mass, $\delta\Omega/\Omega \approx \frac{1}{2}a_0/M_{\text{iso}}(r_c) = 0.72 a_0$: the component’s own mass is exactly where its orbits live.
2. **Narrowing the support helps twice:** the band narrows *and* the shift per unit mass grows (1.55, 1.87 and 2.19 for $\sigma_J = 0.10, 0.05$ and 0.03).
3. **The shift is linear in a_0** over the whole range of the cases: the coefficients computed with `eta.py` are constant to 5 % between $a_0 = 0.008$ and 0.125 .

σ_J	$4\pi\rho_{\text{self}}/a_0$	$M_{\text{self}}/(r_c^3 a_0)$	estimated $\delta\kappa/(\kappa a_0)$	measured $\delta\Omega/(\Omega a_0)$
0.10	9.0×10^{-3}	2.6×10^{-3}	1.15	1.55
0.05	1.28×10^{-2}	2.6×10^{-3}	1.52	1.87
0.03	1.65×10^{-2}	2.6×10^{-3}	1.89	2.19

Table 6: The shift per unit mass for the Gaussian. The uniform-shell estimate reproduces the order of magnitude, the trend (more shift for a narrower shell) and between 74 % and 86 % of the measured value. It is more accurate for narrower shells, where the assumption of nearly circular orbits is better. It underestimates because the Gaussian has many orbits of small amplitude and concentrates the mass near r_c more than a uniform shell. The measured value is the $\delta\Omega$ of Section 1.7.

4. **The shape of F_{eq} matters through the local density.** With $J^2(J_t - J)^2$ the coefficient is 1.81 and 2.08 for $J_t = 0.138$ and 0.083 , close to the Gaussian. The Maxwellian, which concentrates the mass at r_c , gives 2.15, 2.58 and 3.0 for $J_t = 0.276$, 0.138 and 0.083 , and it also *widens* the band: the term $4\pi G\rho(r_c)$ raises $\Omega(0) = \kappa(r_c)$ much more than $\Omega(J_t)$, which belongs to eccentric orbits that spend little time in the shell. With $J_t = 0.138$ the relative width goes from 0.162 to 0.194 at $\eta = 1$; in the hollow families it decreases from 0.166 to 0.136. The same η thus corresponds to different geometries.

3.5 The Landau runs

11_landau has $\eta \approx 0.05$, deep in the damped regime. Its results illustrate the framework.

- **The tail is damped, as expected for $\eta \ll 1$.** The fit of the tail gives $\omega = 0.0570$ and $\gamma = 5.1 \times 10^{-3}$, stable over three time windows, three orders of the fit, and between $N_{rc} = 400$ and 800 .
- **The frequency is inside the band, near its lower edge.** With $a_0 = 10^{-2}$ the band is $[0.0522, 0.0714]$ and $x = 0.25$. This is what the Antonov form predicts: the attractive coupling pushes the pole towards Ω_{min} but is not strong enough to pull it out.
- **Self-gravity changes the *form* of the decay.** Without it the decay of h_1 is Gaussian (Table 4); with it the tail is exponential with rate γ , and at late times the exponential dominates the Gaussian: the ratio between the signals with and without self-gravity goes from 1 to ~ 30 by $t = 1600$.
- **The regime is linear.** The PIC run agrees with linear theory to four digits (0.41998 against 0.41994 at $t = 200$).
- **The numerical floor limits the tail.** The run with $\varepsilon = 0$ gives the noise level, $\approx 10^{-4}$ in the same units. The tail can be followed from $t \approx 1000$ to $t \approx 1600$, about three e -foldings of γ .

The framework then predicts that, as η increases, the mode moves down towards Ω_{min} ($x \rightarrow 0$) and γ decreases; once the mode lies below the band ($x < 0$), γ should be zero within the error. Section 3.7 confirms it.

3.6 Checks of the linear solver

Validation. The case of 11_landau reproduces *bit for bit* the stored linear solution and gives $\omega = 0.0570$, $\gamma = 5.0 \times 10^{-3}$ and $x = 0.25$, as does the fit of the PIC run. The linear solution also gives $|\delta\Phi|_{\text{max}} = 2.3 \times 10^{-4}$ with $\varepsilon = 1$, that is, 2.3×10^{-5} with $\varepsilon = 0.1$, which is exactly what the PIC run measured.

Convergence. In A4 ($\eta = 1$) the pole does not change when Δt is halved, with $N_J = 800$, or with N_Q doubled: $\gamma = 3.66 \times 10^{-4}$ against 3.64×10^{-4} , and ω agrees to five digits. The discrete modes give $|\gamma| \lesssim 10^{-5}$ with either sign, also with Δt halved: this is the floor of the fit, not a growth (the main family cannot grow, by Antonov’s theorem).

The pole does not depend on the test function. $B(J)$ is a choice of the code (Section 1.5); a weight $B(J)e^{-ikQ}$ only needs to be continuous at the circular orbit, which requires B to vanish at least as $J^{k/2}$, and J^2 satisfies this up to $k = 4$. Repeating A4 and L5 with $B \propto \sqrt{J}e^{-(J-0.6J_t)^2/(0.6J_t)^2}$ gives the same ω to four digits and γ to 9% in A4 (3.34 against 3.66×10^{-4}); the fit of $\delta\Phi$, which does not use B , gives 3.14×10^{-4} . γ therefore carries an uncertainty of $\sim 10\%$. The fit of $\delta\Phi$ works only where the pole dominates: at a fixed radius, $\delta\Phi$ carries the algebraic tails of the turning points and of the edge (Section 1.6), which a sum of exponentials cannot represent, and h_1 does not have them precisely because B is smooth and vanishes at $J = 0$.

3.7 The scan in η

case	a_0	η	band $[\Omega_{\min}, \Omega_{\max}]$	ω	x	$\gamma/\Delta\Omega$	γ
L1	0.0030	0.05	[0.0605, 0.0716]		~ 0.6	~ 0.7	no pole
A1	0.0065	0.10	[0.0610, 0.0723]		~ 0.5	~ 0.5	no pole
L2	0.0130	0.20	[0.0618, 0.0735]		~ 0.4	~ 0.4	no pole
A2	0.020	0.30	[0.0627, 0.0749]		~ 0.35	~ 0.3	no pole
L3	0.031	0.45	[0.0641, 0.0770]	0.06673	0.21	0.25	3.3×10^{-3}
A3	0.042	0.59	[0.0655, 0.0791]	0.06738	0.14	0.17	2.4×10^{-3}
L4	0.058	0.80	[0.0676, 0.0823]	0.06857	0.07	0.072	1.1×10^{-3}
A4	0.075	1.00	[0.0698, 0.0857]	0.07023	0.03	0.023	3.7×10^{-4}
L5	0.097	1.24	[0.0727, 0.0902]	0.07247	-0.01	0	$< 10^{-5}$
A5	0.121	1.50	[0.0760, 0.0951]	0.07459	-0.07	0	$< 10^{-5}$
L6	0.170	2.00	[0.0828, 0.1055]	0.07847	-0.19	0	$< 10^{-5}$

Table 7: The main family (Wilson, $W_0 = 3$, $J_t = 0.138$, $L_0 = 2$). For $\eta \leq 0.3$ the damping is so fast ($\gamma/\Delta\Omega \gtrsim 0.3$) that after τ_1 the tail is dominated by phase mixing and there is no isolated pole; the values with $\sim$ are orders of magnitude.

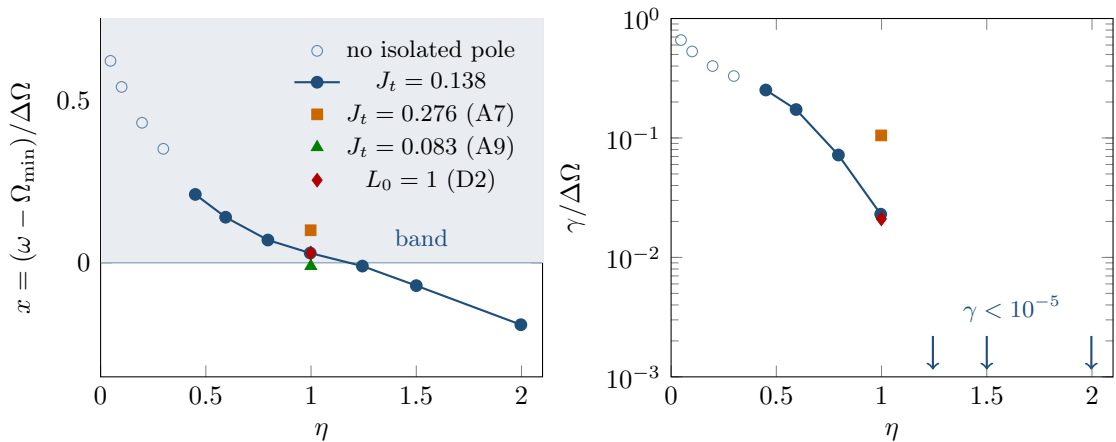


Figure 1: The pole as a function of η (Table 7). Left: position relative to the band (shaded). Right: damping rate; the arrows mark the discrete modes, undamped within the floor of the fit. The isolated markers are the degeneracy test at $\eta = 1$.

There are three results.

1. **The qualitative prediction holds.** As η increases the pole moves down towards $\Omega_{\min}$ and is damped less and less: $\gamma/\Delta\Omega$ falls from 0.25 to 0.023 between $\eta = 0.45$ and 1. Between $\eta = 1.0$ and 1.24 the pole crosses the edge and becomes an undamped discrete mode, increasingly separated from the band ($x = -0.19$ at $\eta = 2$). With $J_t = 0.138$ the transition is at $\eta_c \approx 1.1$ –1.2. In linear theory, self-gravity does sustain an undamped oscillation, with masses of about 10 % of that of the background.
2. **η orders the transition but does not fully collapse it.** Changing L_0 with the width scaled as c changes nothing: D2 agrees with A4 ($x = 0.03$, $\gamma/\Delta\Omega = 0.021$ against 0.023), and D1 with A2. At fixed L_0 , however, the width does matter. At $\eta = 1$ the wide support (A7, $J_t = 0.276$) is still damped ($x = 0.10$, $\gamma/\Delta\Omega = 0.105$), the reference width (A4) is at the edge, and the narrow support (A9, $J_t = 0.083$) already has a discrete mode. At $\eta = 0.3$ the three widths agree within the precision (A2, A6, A8: $x \approx 0.3$ –0.35). The threshold η_c decreases as the support narrows: η is a good design criterion, not the exact variable of the transition. The exact variable is the loop gain at the edge (Section 3.9).
3. **The shape of F_{eq} matters** (Table 8). At $\eta = 1$ the hollow family already has a clear discrete mode ($x = -0.28$); the more thermal Maxwellian ($W_0 = 6$) is still inside the band ($x = 0.13$); and the Gaussian has a pole *below* its 1 % “band” that is *still damped* ($\gamma/\Delta\Omega = 0.020$): without compact support, the orbits of the tail, with a tiny but nonzero F , resonate with it. This shows that the criterion “ ω outside the band” is well defined only with compact support.

case	F_{eq}	a_0	η	x	$\gamma/\Delta\Omega$	result
A4	Wilson, $W_0 = 3$	0.075	1.00	0.03	0.023	damped, at the edge
E3	Wilson, $W_0 = 6$	0.073	1.00	0.13	0.045	damped
E1	hollow, $J^2(J_t - J)^2$	0.075	1.00	-0.28	0	discrete mode
E2	Gaussian, $\sigma_J = 0.05$	0.069	0.97	-0.14	0.020	damped below the 1 % band
H9	hollow, $J_t = 0.083$	0.041	0.99	-0.35	0	discrete mode
H3	hollow, $J_t = 0.083$	0.100	3.1	-1.20	0	discrete mode

Table 8: Different families at $\eta \approx 1$, with $J_t = 0.138$ unless stated otherwise.

3.8 The exponent of the edge

The exponent g of the vacuum edge is that of the dichotomy of Hadžić and collaborators (Section 1.8), and in linear theory it can be scanned directly: for each g we find the mass that gives $\eta = 1$, with $J_t = 0.138$ and $W_0 = 3$, and solve (Table 9).

g	a_0	x	$\gamma/\Delta\Omega$	result at $\eta = 1$
0.75	0.0775	-0.05	0	discrete mode
1 (King)	0.0771	-0.04	0	discrete mode
1.5	0.0762	-0.01	~ 0.004	marginal
2 (A4)	0.075	0.03	0.023	damped
3	0.0739	0.10	0.023	damped

Table 9: The edge at fixed η . At the same η the sharp edges ($g \leq 1$) already have a discrete mode, and the smooth ones ($g \geq 2$) are still damped, more so the smoother the edge.

This is the direction of the dichotomy of Section 1.8: a sharp edge gives oscillations, a smooth edge gives damping. An interpretation, which the calculation supports but does not prove: the

dispersion relation contains $\int F'_{\text{eq}}(J) (\dots) / (\Omega(J) - \omega) dJ$, and near the edge $F'_{\text{eq}} \propto (J_t - J)^{g-1}$. For $g \leq 1$ the integral diverges as $\omega \rightarrow \Omega_{\min}$, so an arbitrarily weak coupling would suffice to pull a mode below the edge, like a bound state at the edge of a band. For $g > 1$ it converges, and a finite coupling is needed: a threshold $\eta_c(g)$ that grows with g . In this sense η_c is a numerical estimate of the $\varepsilon_0(2)$ of the theorem for the isochrone background, expressed in the variable η .

Weak coupling. The direct test is a small η with a sharp edge. $10 \tau_1$ is not enough, so three cases with $\eta = 0.1$ were extended to $60 \tau_1$ (Table 10). For $g \leq 1$ the late signal oscillates exactly at $\Omega_{\min}$ ($x \rightarrow -0.001$) and its decay *slows down*; for $g = 2$ the edge component dies out before $20 \tau_1$, and what remains is the regenerated component at $\Omega_{\max}$, with the slope -2 predicted in Section 1.7. The theorem predicts a mode below $\Omega_{\min}$ for all small masses when $g \leq 1$, which is what is seen at $\eta = 0.1$.

g	late frequency	slope of the envelope of h_1		$ h_1 / h_1(0) $ over 30–60 τ_1
		$7 \rightarrow 14 \tau_1$	$28 \rightarrow 42 \tau_1$	
0.75	$\Omega_{\min}$	−1.35	−0.86	4.0×10^{-4}
1 (King)	$\Omega_{\min}$	−1.71	−1.29	1.3×10^{-4}
2	$\Omega_{\max}$	−2.60	−2.28	6.4×10^{-6}

Table 10: Three cases with $\eta = 0.1$ up to $60 \tau_1$. The slope is $d \ln |h_1|_{\max} / d \ln t$ between consecutive windows; $\delta \Phi$ gives the same (−0.9, −1.3 and −2.2 in the last one).

The time evolution cannot decide, with $60 \tau_1$, whether the response of the sharp edges is a very weakly bound discrete mode or a tail that decays ever more slowly: a mode less than $10^{-3} \Delta \Omega$ below the edge takes $\gtrsim 6 \times 10^5$ to separate from the tail, ten times longer than these runs. What these runs establish is the contrast: with a sharp edge the response at the edge persists, 60 times above that of the smooth edge over 30–60 τ_1 . The loop gain decides the question (Section 3.9): both sharp edges have a discrete mode, $4 \times 10^{-6} \Delta \Omega$ below the edge for $g = 0.75$ and about $10^{-25} \Delta \Omega$ for King, so close that no run can separate it from the response of the edge.

3.9 The loop gain at the edge

`lambda_borde.py` computes the gain $\lambda(\omega)$ of Section 1.7 for each equilibrium. It discretizes $J = J_t(1 - s^2)$ with 200 Gauss–Legendre nodes in s , which removes the square root at the edge, Q with 128 points and the harmonics up to $k = 6$. It takes $r(Q, J)$ from the inverse map of the equilibrium, and does the integrals in Q on a radial grid of 2000 nodes, with linear weights, so that $M = 4\pi L_0 PGP^T$. The mass of the equilibrium is recovered to 10^{-8} . Doubling or halving each resolution changes λ_{edge} of L5 by at most 2×10^{-5} ; keeping only $k \leq 3$ changes it by 4×10^{-4} . Table 11 and Figure 2 give the result for the 25 equilibria with compact support.

There are four results.

1. **The threshold is $\lambda_{\text{edge}} = 1$ for every family.** The 15 damped equilibria have λ_{edge} between 0.047 and 0.881, and the 10 discrete ones between 1.040 and ∞ . The three widths at $\eta = 1$, which η does not separate, have $\lambda_{\text{edge}} = 0.670$ (wide, damped), 0.874 (reference, at the edge) and 1.040 (narrow, discrete). G15, which the time evolution leaves marginal, has $\lambda_{\text{edge}} = 1.051$ and a mode $0.003 \Delta \Omega$ below the edge, consistent with the law $\delta_d \propto (\lambda_{\text{edge}} - 1)^2$ for $g = 3/2$ and the prefactor of Table 12.
2. **The root of $\lambda(\omega) = 1$ is the frequency of the discrete mode.** It agrees with the pole of the linearized solver to 3×10^{-4} or better where that pole is clean. For L5 the value of the table, fitted over 3–9 τ_1 , differs by 4×10^{-4} ; over 10–20 τ_1 the solver gives 0.07250,

case	equilibrium	η	x (linear)	λ_{edge}	ω_d from $\lambda = 1$	ω (linear)
L1	reference	0.05	~ 0.6	0.047	—	—
A1	reference	0.10	~ 0.5	0.100	—	—
L2	reference	0.20	~ 0.4	0.194	—	—
A2	reference	0.30	~ 0.35	0.290	—	—
L3	reference	0.45	0.21	0.428	—	—
A3	reference	0.59	0.14	0.555	—	—
L4	reference	0.80	0.07	0.720	—	—
A4	reference	1.00	0.03	0.874	—	—
L5	reference	1.24	−0.01	1.047	0.07250	0.07247
A5	reference	1.50	−0.07	1.208	0.07458	0.07459
L6	reference	2.00	−0.19	1.468	0.07848	0.07847
A6	$J_t = 0.276$	0.30	~ 0.37	0.243	—	—
A7	$J_t = 0.276$	1.00	0.10	0.670	—	—
A8	$J_t = 0.083$	0.30	~ 0.31	0.328	—	—
A9	$J_t = 0.083$	1.00	−0.01	1.040	0.07076	0.07074
D1	$L_0 = 1$	0.29	~ 0.35	0.288	—	—
D2	$L_0 = 1$	1.00	0.03	0.881	—	—
E3	$W_0 = 6$	1.00	0.13	0.753	—	—
G3	$g = 3$	1.00	0.10	0.772	—	—
G15	$g = 1.5$	1.00	−0.01	1.051	0.06985	marginal
G1	$g = 1$ (King)	1.00	−0.04	∞	0.06945	0.06927
G075	$g = 0.75$	1.00	−0.05	∞	0.06920	0.06911
E1	hollow	1.00	−0.28	3.37	0.06611	0.06609
H9	hollow, $J_t = 0.083$	0.99	−0.35	3.91	0.06767	0.06767
H3	hollow, $J_t = 0.083$	3.10	−1.20	8.02	0.07160	0.07159

Table 11: The loop gain at the lower edge of the band. “Reference” is the Wilson model with $J_t = 0.138$, $W_0 = 3$, $L_0 = 2$ and $g = 2$; the other rows change what is indicated. x and ω come from the linearized solver (Tables 7, 8 and 9); values with $\sim$ have no isolated pole. For $g \leq 1$, λ_{edge} diverges, and its value on the grid (4.1 and 37) only grows with the resolution. In the hollow family F'_{eq} changes sign and the largest real eigenvalue of MR is used; $\lambda(\omega)$ still grows towards the edge.

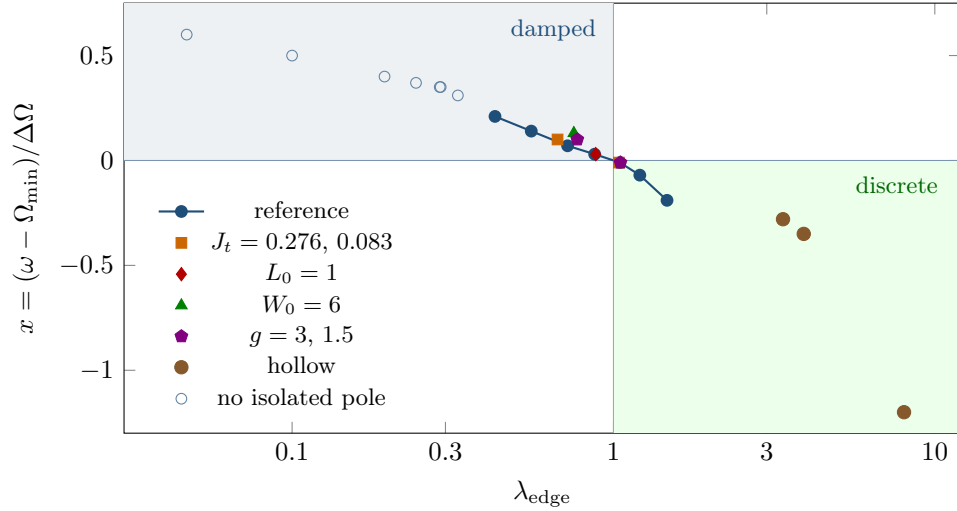


Figure 2: Position of the linear mode against the loop gain at the edge, for all the families of Table 11 except $g \leq 1$, where $\lambda_{\text{edge}} = \infty$. Every damped mode ($x > 0$) has $\lambda_{\text{edge}} < 1$ and every discrete mode ($x < 0$) has $\lambda_{\text{edge}} > 1$: the points occupy only the two shaded quadrants, whatever the width, L_0 , W_0 , g or the family. With η as abscissa (Figure 1) the threshold moves from one family to another.

the same as here. The two calculations are independent: one integrates the initial-value problem, the other solves an eigenvalue problem at fixed frequency. The larger differences, in G1 and G075, are those of fits without a dominant pole.

3. **The critical mass follows directly.** Interpolating λ_{edge} in a_0 , the reference family crosses 1 at $a_0 \approx 0.091$ ($\eta \approx 1.18$) and the narrow support at $a_0 \approx 0.042$. For the wide support $\lambda_{\text{edge}} = 0.670$ at $a_0 = 0.148$, and since $\lambda_{\text{edge}}/a_0$ decreases with the mass, its critical mass is above 0.22. At small mass $\lambda_{\text{edge}} \approx \eta$ in the reference family (0.100 and 0.100 in A1), which is why its η_c is close to 1; the ratio $\lambda_{\text{edge}}/\eta$ falls to 0.88 at $\eta = 1$ and changes with the width (0.81 in A6, 1.09 in A8).
4. **The laws at the edge depend only on g** (Table 12). The local exponent of $\lambda_{\text{edge}} - \lambda(\Omega_{\text{min}} - \delta)$ tends to 1 for $g = 3$, to 0.5 for $g = 3/2$, and for $g = 2$ it grows slowly towards 1, as the logarithm requires ($1 - 1/\ln(1/\delta) \approx 0.87$ at $\delta = 10^{-3}\Delta\Omega$). For $g = 1$, λ grows as $0.15\ln(1/\delta)$.

g	case	λ_{edge}	local exponent for $\delta/\Delta\Omega = 3 \times 10^{-2} \rightarrow 3 \times 10^{-4}$	expected
3	G3	0.772	0.94, 0.97, 0.99, 1.00	1
2	A4	0.874	0.79, 0.84, 0.87, 0.88	1, with a logarithm
1.5	G15	1.051	0.51, 0.51, 0.51, 0.51	0.5
1	G1	∞	$d\lambda/d\ln(1/\delta)$: 0.16, 0.15, 0.15, 0.15	constant

Table 12: How λ approaches its value at the edge. The local exponent is $d\ln[\lambda_{\text{edge}} - \lambda(\Omega_{\text{min}} - \delta)]/d\ln\delta$ between consecutive values of $\delta/\Delta\Omega = 3 \times 10^{-2}$, 10^{-2} , 3×10^{-3} , 10^{-3} and 3×10^{-4} (---borde).

Sharp edges at weak coupling. This settles the question left open in Section 3.8 (Table 13, --debil). With $g \leq 1$ and $\eta = 0.1\text{--}0.3$, λ crosses 1 very close to the edge, so there is a discrete mode, as the theorem says, but bound by a tiny amount. For $g = 0.75$ the distance scales as the

fourth power of the coupling: tripling a_0 (from G075a to G075b) multiplies δ_d by 91, against $3.07^4 = 89$. For King it decreases exponentially: in G1a λ reaches only 0.37 at $\delta = 10^{-8}\Delta\Omega$, and extrapolating the logarithm puts the mode about $10^{-25}\Delta\Omega$ below the edge. Separating such a mode from the response of the edge takes a time of order τ_1/δ_d . That is why the runs of $60\tau_1$ see only a response at $\Omega_{\min}$ that decays ever more slowly.

case	g	η	λ at $\delta/\Delta\Omega =$				$\delta_d/\Delta\Omega$	τ_1/δ_d
			10^{-2}	10^{-4}	10^{-6}	10^{-8}		
G075b	0.75	0.30	0.487	1.373	4.193	13.26	3.9×10^{-4}	$2.6 \times 10^3 \tau_1$
G075a	0.75	0.10	0.167	0.470	1.433	4.417	4.3×10^{-6}	$2.3 \times 10^5 \tau_1$
G1b	1	0.30	0.398	0.621	0.845	1.071	4.3×10^{-8}	$2.3 \times 10^7 \tau_1$
G1a	1	0.10	0.137	0.213	0.289	0.366	$\sim 10^{-25}$	$\sim 10^{25} \tau_1$

Table 13: The sharp edges at weak coupling. δ_d is the distance of the discrete mode below the edge, the root of $\lambda = 1$; for G1a it is an extrapolation of the logarithm.

3.10 Nonlinear scales of the damped cases

The reference run confirms Approximation 9: with $a_0 = 10^{-2}$ and $\varepsilon = 0.1$, $|\delta\Phi|_{\max} = 2.3 \times 10^{-5}$, about fifteen times less than the naive estimate $\varepsilon |\Phi_{\text{self}}|$. In the Landau run, $\omega_b = 1.3 \times 10^{-3}$ with $\varepsilon = 0.1$ and $\gamma = 5.0 \times 10^{-3}$, so $\nu = 0.25$, which is why the PIC run agreed with linear theory to four digits. In the main family with $J_t = 0.138$, as the transition is approached, γ falls to 2.4×10^{-3} , 1.1×10^{-3} and 3.7×10^{-4} at $\eta = 0.6$, 0.8 and 1.0 , while ω_b with $\varepsilon = 1$ hardly changes (2.8 , 3.4 and 4.0×10^{-3}):

$$\nu(\varepsilon) \approx 1.2\sqrt{\varepsilon}, \quad 3.2\sqrt{\varepsilon}, \quad 11\sqrt{\varepsilon}.$$

At $\eta = 1$, $\varepsilon \approx 0.008$ suffices to reach $\nu = 1$ (Table 14).

case		η	γ	ω_b with $\varepsilon = 1$	ν with $\varepsilon = 1$	ε for $\nu = 1$
landau	Gaussian, $\sigma_J = 0.10$	0.05	5.0×10^{-3}	4.0×10^{-3}	0.8	—
A3	Wilson	0.59	2.4×10^{-3}	2.8×10^{-3}	1.2	0.73
L4	Wilson	0.80	1.1×10^{-3}	3.4×10^{-3}	3.2	0.098
A4	Wilson	1.00	3.7×10^{-4}	4.0×10^{-3}	11	0.0084
E3	Wilson, $W_0 = 6$	1.00	8.0×10^{-4}	3.2×10^{-3}	4.0	0.062
A7	Wilson, $J_t = 0.276$	1.00	2.9×10^{-3}	6.0×10^{-3}	2.0	0.24
D2	Wilson, $L_0 = 1$	1.00	1.2×10^{-3}	2.2×10^{-3}	1.9	0.27

Table 14: The two scales of O’Neil’s criterion in the damped cases. ω_b follows Approximation 9, with $\delta\Phi$ at $t = 0$; at the start of the fitting window of the mode it is ~ 0.55 of that value (A4), so ν carries an uncertainty of a factor ~ 2 . In the Landau run, with $\varepsilon = 0.1$, $\nu = 0.25$.

3.11 Questions left for the PIC runs

Linear theory answers the linear questions: where the transition is and how it depends on η , on the width, on the shape and on the edge. It leaves open two nonlinear questions, precisely near the transition.

1. **On the damped side** ($\eta \lesssim \eta_c$), $\gamma \rightarrow 0$ and $\nu = \omega_b/\gamma$ grows without bound for any amplitude, so linear theory fails at late times (Section 1.9). The question is which final state the system selects: damping, saturation or sustained oscillation.

2. **On the discrete side** ($\eta > \eta_c$) there are no resonant particles and O’Neil’s mechanism does not act. The question is whether the mode survives at finite amplitude: its frequency shifts with the amplitude, and a mode just below the edge (L5, $x = -0.01$) can enter the band and be damped, which a separated mode (A5, $x = -0.07$) should not suffer.

4 Design of the PIC battery

4.1 Choice of the distribution

The main family is the Wilson lowered Maxwellian (Section 1.3) with $g = 2$ and $W_0 = 3$, with the perturbation `suave3`. The alternatives are the Gaussian of the Landau runs and the hollow family.

Why not the Gaussian. The Gaussian $J^2 e^{-J^2/\sigma_J^2}$ was proposed for its resemblance to a Maxwell–Boltzmann distribution, but it is not a Maxwellian. In the Maxwell distribution of speeds, $4\pi v^2 e^{-v^2/2\sigma^2}$, the factor v^2 is the Jacobian of the velocity shell; the distribution function itself is $e^{-v^2/2\sigma^2}$, maximal at $v = 0$. Here the measure is already flat, $dr dp_r = dQ dJ$, so the J^2 of F is not a Jacobian but a genuine depletion of phase space around the circular orbit. At $r = r_c$, where $J \approx p_r^2/2\kappa$ (Section 1.2), it gives a distribution of p_r proportional to $p_r^4 e^{-p_r^4/4\kappa^2\sigma_J^2}$: zero at $p_r = 0$ and with two maxima, two crossing radial streams. It also has a population inversion, $\partial F/\partial E > 0$ for $J < \sigma_J$, so Antonov’s theorem does not guarantee its stability.

The correct analogue. $F \propto e^{-E/T}$. Since $E = p_r^2/2 + \Phi_{\text{ef}}(r)$, at each radius it gives *exactly* a Gaussian in p_r with dispersion $\sqrt{T}$ and a Boltzmann density $e^{-\Phi_{\text{ef}}/T}$, and since $E = E(J)$ it is an exact equilibrium. Near the circular orbit $E \approx E_c + \kappa J$, and $F \propto e^{-\kappa J/\sigma_R^2}$, the quasi-isothermal distribution of Binney (2010, MNRAS 401, 2318), standard for warm discs in action variables. The width has a physical meaning, $J_0 = \sigma_R^2/\kappa$, with $\sigma_R = \sqrt{T}$ the radial velocity dispersion; with $W_0 = 3$ and $J_t = 0.138$, $\sigma_R/v_c = 0.16$ relative to the circular velocity $v_c = L_0/r_c$: a warm shell.

Compact support from tides. A cluster in an external field loses the stars with $E > E_t$. The family of lowered isothermal models of Gieles and Zocchi (2015, MNRAS 454, 576), which is $E_\gamma(g, x)$, describes this cut: King (1966) with $g = 1$ and Wilson (1975) with $g = 2$. Its exponent g is the one of the Hadžić dichotomy, and the family contains as limits the polytrope, $(E_t - E)^g$, when $W_0 \rightarrow 0$, and the isothermal sphere when $W_0 \rightarrow \infty$.

	Gaussian $J^2 e^{-J^2/\sigma_J^2}$	hollow $J^2 (J_t - J)^2$	Wilson $E_\gamma(2, (E_t - E)/T)$
compact support	no	yes	yes, from tides
band	1 % threshold	exact	exact
distribution of p_r at r_c	bimodal	bimodal	Gaussian (lowered)
at $J = 0$	$\propto J^2$	$\propto J^2$	maximum
monotonic in E (Antonov)	no	no	yes
smooth perturbation	$s = 1$	$s = 1$	$s = (J/J_t)^{3/2}$
free tail of $\delta\Phi$ at $10\tau_1$	1.2×10^{-4}	4.2×10^{-4}	3.0×10^{-4}
ρ_Φ	1.00	1.09	0.11
a_0 for $\eta = 1$	0.069	0.075	0.075
$\delta\Omega/(\Omega a_0)$	1.87	1.81	2.58
band when mass is added	narrows	narrows	widens

Table 15: The three families with the reference width ($\sigma_J = 0.05$, $J_t = 0.138$, $L_0 = 2$). Tails and ρ_Φ from Table 5; masses and shift coefficients from `eta.py`.

The reasons for the choice, in order of importance, are the following.

1. **It is monotonic**, $\partial F/\partial E < 0$: by Antonov's theorem it has no growing radial modes (Section 1.7), and an oscillation that does not decay is a genuine neutral mode, without the ambiguity of an instability. It also satisfies the monotonicity hypothesis under which the theory of Hadžić and collaborators is formulated; the two hollow families do not.
2. **Its interior is thermal**: at each radius the distribution of p_r is Gaussian, lowered near E_t . With $W_0 = 3$ the maximum is 80 % of that of the untruncated isothermal model, $P(2, 3) = 0.80$. Its parameters have a physical meaning: temperature, tidal energy and the shape of the edge.
3. **Its compact support has a physical cause**, the tides, and it leaves the band exact: the essential spectrum and η are properties of the equilibrium, not of a threshold.
4. **Its edge is the one of the Hadžić dichotomy, and it is adjustable**: $g = 2$ is in the damped class, and the edge experiment is the same distribution with another g , including King ($g = 1$), the marginal case.

Why $g = 2$ and $W_0 = 3$. We explored $g \in \{2, 3\}$ and $W_0 \in \{1, \dots, 8\}$ with $J_t = 0.138$ (Table 16). All give $\eta = 1$ with a_0 between 0.066 and 0.074; what changes is the shape. Raising W_0 makes the core more thermal, but it leaves the edge without weight and weakens the smooth perturbation; at the same W_0 , $g = 2$ has a more thermal core, a lower tail and more potential than $g = 3$. $W_0 = 3$ is the compromise: a core at 80 % of the isothermal one, and 7 % of the mass in the outer half of the support, which is where the edge lives. $W_0 = 6$, the nearly thermal limit, enters as case E3 and run V2.

g	W_0	$P(g, W_0)$	$\langle J \rangle/J_t$	mass at $J > J_t/2$	σ_R/v_c	free tail at $10\tau_1$	ρ_Φ
2	3	0.80	0.20	0.068	0.157	3.0×10^{-4}	0.11
2	6	0.98	0.15	0.028	0.111	1.0×10^{-3}	0.06
3	3	0.58	0.17	0.037	0.157	6.8×10^{-4}	0.08
3	5	0.88	0.15	0.024	0.122	1.0×10^{-3}	0.06

Table 16: Candidates of the Wilson family, with $J_t = 0.138$, $L_0 = 2$ and $s = (J/J_t)^{3/2}$. $P(g, W_0) = E_\gamma(g, W_0)/e^{W_0}$ is the ratio between the maximum and that of the untruncated isothermal model. The full scan in W_0 (1, 2, 3, 5, 8) shows the same trend: $\langle J \rangle/J_t$ decreases from 0.23 to 0.12 and the outer mass from 10 % to 1 %.

The cost: the edge at $\Omega_{\max}$. Every monotonic distribution is maximal at $J = 0$, so the upper edge of the band, $\Omega_{\max} = \kappa(r_c)$, carries weight. This has three consequences.

- (a) The flat perturbation is discontinuous at the circular orbit (Section 1.4), and it leaves in $\delta\Phi$ a tail $t^{-3/2}$ at $\Omega_{\max}$, 150 times above the Gaussian at $10\tau_1$. With $s = (J/J_t)^{3/2}$ the tail drops to the level of the Gaussian (Table 5).
- (b) That smooth perturbation produces little potential, $\rho_\Phi = 0.11$: for the same $\delta\Phi$ it needs an ε nine times larger than the Gaussian. This does not limit the nonlinear study, because near the transition the controlling parameter is $\nu = \omega_b/\gamma$, which is reached with $\varepsilon \ll 1$ (Section 1.9).
- (c) Self-gravity regenerates a component at $\Omega_{\max}$ with tail t^{-2} (Section 1.7) that no initial condition removes. It is fitted as a separate component, with known frequency and exponent, and its amplitude is computed beforehand with `lineal.py`.

In addition, the band widens with the mass (Section 3.4), so the same η describes a different geometry than in the hollow families. Case E1 measures whether this matters, and it does (Section 3.7).

Why not the pure polytrope. $(E_t - E)^g$ is the limit $W_0 \rightarrow 0$ of this family: it keeps the three costs above and loses the thermal interior.

4.2 Numerical decisions

The angular momentum L_0 . L_0 decides where the system lives inside the isochrone, and with it how much self-gravity is obtained per unit mass (Table 17).

L_0	c	circular radius r_c	radial period	$M_{\text{iso}}(r_c)$
0.25	1.133	0.804	9.1	0.078
0.5	1.281	1.30	13.2	0.192
1	1.618	2.42	26.6	0.413
2	2.414	5.74	88.4	0.696
3	3.303	10.9	226	0.828

Table 17: With $L_0 = 2$ the distribution lives where the background already encloses 70 % of its mass, so a_0 competes against 0.7. With $L_0 = 1$ it competes against 0.41: the same η is reached with less mass.

The cost per run is almost independent of L_0 : the number of steps needed to cover the same number of mixing times scales as T_r/r_c , which is 15.3, 11.0 and 11.3 for $L_0 = 2, 1$ and 0.25. We take $L_0 = 2$ as the reference, for continuity with the runs, equilibria and analyses already validated, and $L_0 = 1$ as a control, with the linear cases D1 and D2 and the PIC run V1. If the result collapses in η for two values of L_0 , the claim no longer depends on that choice: L_0 changes at once the enclosed background mass, the radius, the period and the shape of $\Omega(J)$.

A fair comparison requires scaling the width, because what matters is $\Delta\Omega/\Omega \approx 3\Delta J/c$: **the width of the support must scale as c .** The equivalent of $J_t = 0.138$ with $L_0 = 2$ is $J_t = 0.094$ with $L_0 = 1$ ($\sigma_J = 0.034$). The grid must also scale with r_c : with $L_0 = 1$ the system lives at $r \sim 1.5$ –4, so $\Delta r = 0.05$ and $r_{\text{max}} = 8$ give the same relative resolution as $\Delta r = 0.1$ and $r_{\text{max}} = 20$ with $L_0 = 2$.

Why vary the width and not only the mass. With the width of the Landau runs ($J_t = 0.276$, or $\sigma_J = 0.10$) the shift is linear in the mass, $\delta\Omega/\Omega \approx 2.1a_0$ for the Maxwellian, while the width of the band is ≈ 0.31 . The masses 10^{-3} , 10^{-2} and 10^{-1} then give $\eta \approx 0.007$, 0.07 and 0.68, *all three on the same side*. Such a scan would measure damping in all three cases and would not distinguish between “self-gravity is not enough” and “the regime was not reached”. With $J_t = 0.083$ the band without mass shrinks to 0.10, and the mass $a_0 = 0.044$ already gives $\eta \approx 1.0$: **the width of the support is the parameter that allows the transition to be crossed at small masses.**

One could instead raise a_0 up to 0.3 or 0.5 at $J_t = 0.276$, and the transition would also be crossed. We do not, for three reasons: (i) it mixes two changes, since raising the mass shifts the frequencies *and* deforms the equilibrium, so the transition cannot be attributed to a single cause; (ii) it leaves the stated regime, because with $a_0 = 0.5$ the self-gravitating component is comparable to the enclosed background mass; and (iii) it increases the risk of instability (Section 4.5).

Duration of the runs and the recurrence ceiling. Two scales set the duration. The first is phase mixing: the linear runs last $t_{\text{fin}} = 10 \tau_1$ (Section 3.3), and the nonlinear runs also several bounce periods (Section 4.3).

The second is *recurrence*, an artefact of the regular grid in J . Two neighbouring nodes, separated by $\Delta J = J_{\text{max}}/N_{rc}$, have frequencies that differ by $|d\Omega/dJ| \Delta J$, and their harmonic k accumulates a phase difference $k |d\Omega/dJ| \Delta J t$. When this difference reaches 2π , all pairs of neighbours are in phase again: the grid cannot represent structure finer than ΔJ , and phase mixing undoes itself. This happens at

$$T_{\text{rec}}(k) = \frac{2\pi}{k |d\Omega/dJ| \Delta J}.$$

Beyond T_{rec} any measured plateau is an artefact, and the limit is *different for each k* : it decreases as $1/k$. This is the effect of Birdsall and Langdon seen from the other side.

With compact support and $J_{\text{max}} = J_t$ the count simplifies: the mean slope is $|d\Omega/dJ| \approx \Delta\Omega/J_t$ and $\Delta J = J_t/N_{rc}$, so

$$T_{\text{rec}}(k) = \frac{N_{rc}}{k} \tau_1,$$

independent of the width and of the mass. With $N_{rc} = 400$, $T_{\text{rec}}(1) = 400 \tau_1$, which in A4 is 1.6×10^5 , above the longest run (5.2×10^4). For $k = 4$ the ceiling drops to $100 \tau_1 \approx 4 \times 10^4$, and in N6 and in its reference with $\varepsilon = 0$ one must state up to when h_4 is valid. For the Gaussian (case E2), $J_{\text{max}} = 6\sigma_J$ covers 2.17 times the support and the margin shrinks by that factor. This is why J_{max} must follow the support and not stay at its historical value of 0.6, which with $\sigma_J = 0.03$ would waste most nodes where $F \approx 0$. The generator uses by default $J_{\text{max}} = J_t$ for the compact families and $6\sigma_J$ for the Gaussian.

Resolution. There are four independent sources of error, and they do not substitute for one another:

- N_{rc} (nodes in J) sets the time window before recurrence and how many nodes a trapping island covers;
- N_{pc} (nodes in Q) sets how many modes k are reliable; convergence in Q is spectral, and 25 resolves well up to $k = 4$;
- Δr sets the error of the self-gravity (deposition and Poisson solver);
- Δt sets the error of the time integrator.

4.3 The runs

The PIC runs are used where linear theory fails: near the transition, at late times. Each run has as reference the linear solution of its own equilibrium (Section 4.4), and what is measured is the deviation from it.

The common configuration, unless stated otherwise, is the main family (Wilson, $g = 2$, $W_0 = 3$) with $J_{\text{max}} = J_t$, $L_0 = 2$, a grid $r \in [0, 20]$ with $\Delta r = 0.1$, **courant** = 1 ($\Delta t = 0.05$), yoshida4, $N_{rc} = 400$, $N_{pc} = 25$ ($N = 10^4$), $\chi(Q) = \cos Q$, the perturbation **suave3** and the test function $j_1 = s_{j1} = J_t/2.763$. The amplitude ε of each point is chosen for a given ν : since $\omega_b \propto \sqrt{\varepsilon}$, $\varepsilon = (\nu\gamma/\omega_b^{(1)})^2$, with γ and $\omega_b^{(1)}$ (the value at $\varepsilon = 1$) from block L (Table 14). The runs with $\nu \geq 1$ last at least three bounce periods, $3 \cdot 2\pi/\omega_b$; those with $\nu < 1$ and those with discrete modes last 10 and $20 \tau_1$.

#	base	(η, x)	ν	ε	ω_b	$2\pi/\omega_b$	$\Delta J_{\text{atr}}/\Delta J$	t_{fin}	cost
N1	A3	(0.59, 0.14)	0.3	0.065	7.1×10^{-4}	8800	42	4600	2.6 min
N2	A3		1	0.73	2.4×10^{-3}	2700	139	8000	4.5 min
N3	L4	(0.80, 0.07)	1	0.098	1.1×10^{-3}	5900	58	17800	10.0 min
N4	L4		3	0.88	3.2×10^{-3}	2000	173	5900	3.3 min
N5	A4	(1.00, 0.03)	0.3	0.00075	1.1×10^{-4}	57000	6	3950	2.2 min
N6	A4		1	0.0084	3.7×10^{-4}	17000	18	51500	28.7 min
N7	A4		3	0.075	1.1×10^{-3}	5700	55	17200	9.6 min
N8	A4		10	0.84	3.7×10^{-3}	1700	184	5200	2.9 min
N9	A5	(1.50, -0.07)	–	0.1	1.8×10^{-3}	3600	74	6600	3.6 min
N10	A5		–	1.0	5.6×10^{-3}	1100	233	6600	3.6 min
N11	L5	(1.24, -0.01)	–	0.1	1.5×10^{-3}	4200	69	7200	4.0 min
N12	L5		–	1.0	4.8×10^{-3}	1300	218	7200	4.0 min

Table 18: N1–N8 cover the damped side at three distances from the transition ($\eta = 0.59, 0.80$ and 1.00) with ν from 0.3 to 10; since γ falls as η_c is approached, the same ν is reached with ever smaller ε . N1 and N5 ($\nu = 0.3$) should agree with linear theory: they are the validation. N9–N12 are the discrete side: a mode separated from the edge (A5) and one just below it (L5), at small and large amplitude; there ω_b is only a measure of the amplitude, since there are no resonant particles. $\Delta J_{\text{atr}}/\Delta J$ is the half-width of the island in nodes of J ; with $\mu = \omega_b/\Delta\Omega \leq 0.3$ in the whole block, the continuum as a whole is not trapped.

Block N: the nonlinear map. Table 18.

The block tests O’Neil’s predictions for the damped side: for $\nu \ll 1$, linear damping at all times; for $\nu \sim 1$, the amplitude stops falling near $t \sim 2\pi/\omega_b$ and saturates at a fraction of its initial value; for $\nu \gg 1$, it hardly decays, and the envelope oscillates at a frequency $\sim \omega_b$. The separation time t_{nl} should scale as $\varepsilon^{-1/2}$ (N5, N6, N7 and N8 at fixed η).

Block V: what the nonlinear behaviour depends on. Four runs with the same ν as a point of block N, on another equilibrium, test whether the final state is fixed by (η, ν) or also by the width, L_0 and the shape of the distribution (Table 19).

#	base	ν	ε	$2\pi/\omega_b$	t_{fin}	cost	compare with
V1	D2 ($L_0 = 1$, equal to A4 in linear theory)	1	0.27	5400	16300	17.0 min	N6
V2	E3 (Wilson, $W_0 = 6$)	3	0.56	2600	7800	4.5 min	N7
V3	A7 ($J_t = 0.276$)	1	0.24	2100	6400	3.4 min	N6, N2
V4	E1 (hollow, discrete mode)	–	1.0	600	12400	7.0 min	N10

Table 19: V1 isolates L_0 : in linear theory D2 and A4 are equal in units of the band. V2 and V3 change the shape and the width at fixed ν . V4 is the discrete mode of a non-monotonic family at large amplitude ($\mu \approx 1$): if it grows, it is an instability (Antonov’s theorem does not exclude it there). V1 costs more because with $L_0 = 1$ the grid is finer ($\Delta r = 0.05$, $r_{\text{max}} = 8$) and Δt is halved.

Block Z: the unperturbed reference. As in 11_landau, each equilibrium is also run with $\varepsilon = 0$ and the same duration, and the observable is the difference $[X(\varepsilon) - X(0)]/\varepsilon$. This subtracts the evolution of the discrete equilibrium itself (noise and numerical drift), which is not negligible in the long runs with small ε . There are nine runs, one per equilibrium used (A3, L4, A4, A5, L5, D2, E3, A7, E1), each with the duration of its longest run: 83 min in total, almost all in A4 (51500 time units, 29 min).

Block B: convergence. On N7 (A4, $\nu = 3$), changing one parameter at a time, and the noise floor on the point with the smallest signal (Table 20).

#	change	what it tests	cost
B1	$N_{rc} = 800$ ($N = 2 \times 10^4$)	resolution of the island in J	19 min
B2	$N_{pc} = 50$	harmonics k generated by the nonlinearity	19 min
B3	$\Delta r = 0.05$, courant = 2	error of the self-gravity at fixed Δt	10 min
B4	courant = 0.5	error of the integrator	19 min
B5	N5 with $N = 5 \times 10^4$	noise floor with $\varepsilon = 7.5 \times 10^{-4}$	11 min

Table 20: Acceptance criterion: the envelope of $\delta\Phi$, t_{nl} and the saturation amplitude must change by less than 5 % when each parameter is refined. Each parameter is refined in one direction only, to limit the cost; if one of them fails, the next refinement step is added.

Cost. *Approximation 13:* the cost is taken as linear in N and in the number of steps, calibrated with an actual run (Table 21).

block	purpose	runs	cost
L	linear theory (computed)	31 + 12	done
N	nonlinear map in (η, ν) and discrete mode	12	79 min
V	variations: L_0 , shape, width, hollow family	4	32 min
Z	references with $\varepsilon = 0$	9	83 min
B	convergence and noise floor	5	78 min
PIC total		30	≈ 4.5 h
minimal core: N5–N10, Z of A4 and A5, B1, B4		12	≈ 2.0 h

Table 21: With four threads, calibrated with an actual run (100 s for 10^4 particles, 6×10^4 steps, yoshida4 and self-gravity). The linear reference of each long run adds about 40 min, without particles.

4.4 Measurement protocol

Linear reference. Before each PIC run the linear problem is solved on the same equilibrium (`lineal.py`, or `barrido_lineal.py` for the cases of this document). It gives the linear h_k and $\delta\Phi$ for $\varepsilon = 1$, the pole (ω, γ) , the regenerated component at $\Omega_{\max}$ and ω_b . In 11_landau the PIC run agreed with it to four digits.

Observables of each PIC run.

1. **The deviation from linear theory**, $D(t) = \|\delta\Phi_{\text{PIC}} - \varepsilon \delta\Phi_{\text{lin}}\| / \|\varepsilon \delta\Phi_{\text{lin}}\|$, with $\delta\Phi$ the perturbation of the potential with respect to the equilibrium (`landau_analysis.py`). *It does not depend on any angle-action map.* The time t_{nl} at which D crosses 0.1 is compared with $2\pi/\omega_b$: O’Neil predicts $t_{nl} \propto \varepsilon^{-1/2}$.
2. **The late envelope of $\delta\Phi$** : damping (linear), saturation (the ratio A_{sat}/A_0 between the final and initial amplitudes) and oscillation of the envelope, whose frequency is compared with ω_b . For the main family the fit also includes the component at $\Omega_{\max}$, with its amplitude taken from the linear reference.

3. **The plateau of F at the resonance:** the average over Q of F around J_r , with $\Omega(J_r) = \omega$, should flatten over a width $\sim \Delta J_{\text{atr}}$. It is the direct signature of trapping, and it requires mapping the particles to the angle-action frame of the equilibrium (`aa_numerico.py`).
4. $h_k(t)$ for $k = 0 \dots 4$ (Section 1.5): a frequency common to all k is the signature of a mode; the generation of $k = 2$ with amplitude $\propto \varepsilon^2$ is the signature of the nonlinearity.
5. **The drift of J and energy conservation,** which bound the time window in which the result belongs to the system and not to the method.

observation	interpretation	confirming control
PIC equal to linear at all t departs at $t \sim 2\pi/\omega_b$, the envelope oscillates and saturates	linear regime, $\nu \ll 1$ trapping (O’Neil)	$D < 10^{-2}$; scales with ε $t_{\text{nl}} \propto \varepsilon^{-1/2}$; plateau of F of width ΔJ_{atr} at J_r
constant amplitude, common ω outside the band	discrete mode	also in the linear solution
exponential growth	instability	also grows at smaller ε ; excluded in the main family
flat plateau at the level of the floor	discretization	decreases as N increases

Table 22: How each possible result is read, and which control distinguishes it from the others.

4.5 Risks and controls

1. **Instability instead of oscillation.** The main family is monotonic and, by Antonov’s theorem, has no growing radial modes. The hollow family and the Gaussian do increase between $J = 0$ and their maximum ($\partial F/\partial E > 0$ there): in them a growth is reported as an instability, not as “no damping”.
2. **Discretization floor.** The points with $\nu \approx 1$ use a small ε and a small signal. The floor must lie clearly below the observed amplitude for several bounce periods; this is the role of run B5, with $N = 5 \times 10^4$ on the point with the smallest signal.
3. **Resolution of the island.** The half-width $\Delta J_{\text{atr}} = 2\omega_b/|\Omega'|$ must cover many nodes in J : Table 18 gives the ratio $\Delta J_{\text{atr}}/\Delta J$ for each point.
4. **Recurrence.** Any plateau measured beyond $T_{\text{rec}}(k) = N_{rc}\tau_1/k$ is an artefact of the grid in J . The long runs of block N stay below $T_{\text{rec}}(1)$; for $k = 4$ one must state up to which t it holds.
5. **Numerical drift of the equilibrium.** If J drifts appreciably within the window, what is measured is numerical relaxation and not trapping. This is the actual limit of the useful window of the long runs, and it is controlled with $\varepsilon = 0$ on the same equilibrium.
6. **Nearly circular nodes.** In the main family the nodes with $J \rightarrow 0$ have the largest weights, and the map is degenerate there. The guards on the radicands (item E13 of the audit) cover the case, but each run must be checked for discarded particles.
7. **Edge tail in the main family.** The component at Ω_{max} decays as t^{-2} ; a single-term fit would mistake it for a slow damping. It is fitted separately, with its frequency fixed at the edge and its amplitude from the linear reference.

5 Summary

5.1 Approximations

Table 23 lists the approximations in the order in which they appear, with the section where each is introduced and how it was checked.

#	approximation	where it is used	validation
1	Epicycle near the circular orbit (1.2)	regularity at $J = 0$, $C \propto \sqrt{J}$, R_e	exact in the circular limit: $\kappa = \Omega(0) = 0.0711$
2	1 % threshold for the Gaussian (1.6)	edges of the band, η	unnecessary with compact support; a convention common to the Gaussian cases
3	$\Delta\Omega/\Omega \approx 3\Delta J/(J + c)$ (1.6)	reading of the band	error < 1 % against the exact isochrone
4	τ_k as a scale (1.6)	duration of the runs	measured free decay: Gaussian, $\Delta\Omega/\sigma_\Omega \approx 6.2\text{--}7.0$
5	Uniform shell (1.7)	estimate of the shift	reproduces the trend and 74 %–86 % of the value; the computed value is used
6	Regenerated component at $\Omega_{\max}$ (1.7)	reading of the tail	confirmed in linear theory: slope -2.2 to -2.3 in A1 up to $60\tau_1$
7	The shift measures the coupling (1.7)	definition of η	heuristic; η orders the transition but does not fix it alone; $\lambda_{\text{edge}} = 1$ does (3.9)
8	Pendulum, isolated resonance (1.9)	ω_b , ΔJ_{atr} , μ , ν	standard in nonlinear dynamics
9	$\omega_b \approx \sqrt{ \Omega' \max_r \delta\Phi }$, $ \Omega' \approx \Delta\Omega/J_t$ (1.9)	nonlinear map	$\delta\Phi$ from the linear solution of each case; ν within a factor ~ 2
10	O’Neil’s criterion (1.9)	linear/nonlinear boundary	a plasma result (one wave, 1D); it is what the PIC runs test
11	Dominant pole by matrix pencil (2.4)	ω , γ , x of block L	reproduces the pole of 11_landau; independent of B to 10 %
12	Tails in the bare isochrone (2.5)	choice of F_{eq} and s	exact quadrature; reproduces <code>landau_libre</code> ; exponents confirmed
13	Cost linear in N and in the number of steps (4.3)	times	calibrated with $N = 10^4$; optimistic for $N = 5 \times 10^4$ if it scales worse in parallel

Table 23: The approximations of the document, in order of appearance, with the section where each is introduced and how it was checked.

5.2 Scope

The design does not cover the following.

- **Initial conditions far from an equilibrium**, such as the general function of Q of the draft or a cold collapse. There is no F_{eq} about which to linearize, and the final state is not known in advance: it is the natural territory of the code, but it requires its own design.
- **The point-mass background** of the work of Hadžić and collaborators. The code can reproduce it (`BGtype = sphere` is $-1/r$ outside unit radius), but the numerical angle–action map is written around the isochrone, and the background would have to be parametrized.
- **Distributed angular momentum**. The edge at $\Omega_{\max}$ and its tail t^{-2} are specific to $\delta(L - L_0)$; with a distribution in L , $\kappa(L)$ varies and blurs them.

- **The regime of comparable masses** ($a_0 \gtrsim 0.3$), a legitimate but different physical problem.

A Notation

symbol	meaning	section
$f, F(r, p_r)$	full distribution and distribution reduced at fixed angular momentum	1.1
L_0	common angular momentum of all particles	1.1
$\Phi_{\text{iso}}, \Phi_{\text{self}}, \Phi_{\text{ef}}$	background, self-consistent and effective potential (with the centrifugal barrier)	1.1
(Q, J)	radial angle and action	1.2
$\Omega(J), T_r(J)$	radial frequency and period	1.2
c	isochrone constant, $c = \frac{1}{2}(L_0 + \sqrt{L_0^2 + 4})$	1.2
r_c, κ, E_c	radius, epicyclic frequency and energy of the circular orbit	1.2
$R_e(J)$	epicycle radius, $\sqrt{2J/\kappa}$	1.2
$F_{\text{eq}}(J), A$	equilibrium and its amplitude	1.3
a_0	mass of the self-gravitating component	1.3
σ_J	width of the Gaussian distribution	1.3
J_t	edge of the support in action	1.3
g	exponent of the vacuum edge, $F_{\text{eq}} \propto (J_t - J)^g$ (the k of Hadžić)	1.3
E_t, T, W_0	tidal energy, temperature and depth of the lowered Maxwellian	1.3
$\varepsilon, \chi(Q), s(J)$	amplitude, angular profile and radial factor of the perturbation	1.4
N_{rc}, N_{pc}, N	nodes in J , nodes in Q and number of particles	2.2
$\delta F, \delta \Phi$	perturbation of the distribution and of the potential	1.5
$\ \delta \Phi\ $	rms value of $\delta \Phi$ over $r \in [3, 15]$	1.5
$h_k, B(J), j_1, s_{j1}$	harmonic k of the distribution and its test function	1.5
$\beta(Q, J), \beta_{\pm 1}(J)$	weight of a linear observable and its coefficients $k = \pm 1$	1.5
$\Omega_{\text{min}}, \Omega_{\text{max}}, \Delta \Omega$	edges and width of the frequency band	1.6
$\bar{\Omega}$	mass-weighted mean frequency	1.6
$\bar{\Omega}_{\text{iso}}$	the same, for the same distribution without self-gravity ($a_0 \rightarrow 0$)	1.7
τ_k	mixing time of harmonic k , $2\pi/(k \Delta \Omega)$	1.6
$w(\Omega)$	distribution of frequencies seen by an observable	1.6
ρ_Φ	$\delta \Phi(0)$ per unit mass and unit ε , relative to the Gaussian	3.3
ω, γ	frequency and damping rate of the collective mode	1.7
x	position of the mode in the band, $(\omega - \Omega_{\text{min}})/\Delta \Omega$	1.7
$\delta \Omega, \eta$	relative shift of $\bar{\Omega}$ and design criterion	1.7
$\omega_b, \Delta J_{\text{atr}}$	bounce frequency and half-width of the trapping island	1.9
μ, ν	$\omega_b/\Delta \Omega$ and ω_b/γ	1.9
η_c	value of η at the transition	3.7
$\lambda(\omega), \lambda_{\text{edge}}$	gain of the self-consistent loop, and its value at Ω_{min}	1.7
δ_d	distance of a discrete mode below the edge, in units of $\Delta \Omega$	3.9
$T_{\text{rec}}(k)$	recurrence time of the grid in J	4.2
t_{fin}	duration of a run	4.2

Table 24: Symbols and the section where each is defined. k is always a Fourier index in Q ; the exponent of the edge is called g in all families, although Hadžić and collaborators call it k .

B Tools

The changes to the scripts are in the repository and are neutral: with their default values they reproduce the previous results bit for bit (initial state of the Landau run, linear run 400×25 ,

polytrope initial states and tables of `eta.py`). The only output that changed is the T_{rec} column of `eta.py`: $|d\Omega/dJ|$ was obtained by differentiating the table $E(J)$ twice, which gave noise of order one between neighbouring nodes, and it is now the mean slope over the support.

- `equilibrio.py`: the option `--forma` selects `gauss` (`--sigma`), `politropo` ($J^m(J_t - J)^k$, with `--jt`, `--k` and `--m`) or `maxwell` ($E_\gamma(k, (E_t - E)/T)$, with `--jt`, `--k` and `--w0`); the other options are `--jmax`, `--l0`, `--gq` and `--pert` (`plana`, `suave` or `suave3`). It checks the positivity of F and stores in the equilibrium file the shape, (J_t, k, m, W_0) and the perturbation. E_γ is evaluated with its series of positive terms.
- `lineal.py`: the linearized Vlasov–Poisson solver; it takes the shape and the perturbation from the equilibrium; `--j1` `--sj1` `--pb` set the test function $B(J) = J^{p_b} e^{-(J-j_1)^2/s_{j_1}^2}$ ($p_b = 2$, the one of the code, by default).
- `barrido_lineal.py`: block L. For each case it builds the equilibrium, solves the linear dynamics, fits the pole with matrix pencil in windows $[a, 2a]$ τ_1 and gives ω_b . It is validated by reproducing bit for bit the linear solution of `11_landau` and its pole.
- `eta.py`: band, shift, η , τ_1 , T_{rec} and cost of a run, without simulating.
- `lambda_borde.py`: the loop gain $\lambda(\omega)$, λ_{edge} and the frequency of the discrete mode of each equilibrium (Section 3.9); with `--borde`, the approach to the edge, and with `--debil`, the sharp edges at weak coupling. All of it takes 11 min and is written to `exe/eta_lineal/lambda_borde.txt`.
- `colas_libres.py`: exact free decay of h_1 and $\delta\Phi$ by quadrature, and ρ_Φ .
- `landau_analisis.py`: $\delta\Phi$ and h_k of the PIC runs in the frame of the equilibrium.

Example of use:

```
python3 barrido_lineal.py --casos A4 E1 G1
python3 equilibrio.py --forma maxwell --jt 0.138 --k 2 --w0 3 --a0 0.075 \
    --eps 0.07 --pert suave3 --nrc 400 --npc 25 --salida eta/N2.dat
python3 eta.py --forma maxwell --jt 0.138 --k 2 --w0 3 --a0 0.075
python3 colas_libres.py
```